

SECTION TABLE OF CONTENTS

DIVISION 01 - GENERAL REQUIREMENTS

SECTION 01 11 00

SUMMARY OF WORK

08/15, CHG 2: 08/21

PART 1 GENERAL

- 1.1 SUBMITTALS
- 1.2 WORK COVERED BY CONTRACT DOCUMENTS
 - 1.2.1 Project Description
 - 1.2.2 Location
- 1.3 CONTRACT DRAWINGS
- 1.4 LOCATION OF UNDERGROUND UTILITIES
 - 1.4.1 Notification Prior to Excavation
- 1.5 SALVAGE MATERIAL AND EQUIPMENT

PART 2 PRODUCTS

PART 3 EXECUTION

-- End of Section Table of Contents --

SECTION 01 11 00

SUMMARY OF WORK
08/15, CHG 2: 08/21

PART 1 GENERAL

1.1 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals
Salvage Plan; G

1.2 WORK COVERED BY CONTRACT DOCUMENTS

1.2.1 Project Description

This work replaces existing aged fuel piping from bulk storage (ML-2) to Pier #3 with a new 20-inch transfer pipeline (approx. pipeline length 9350 LF). The piers are currently limited in flow and/or concurrent delivery due to existing main line characteristics. The pipeline will be routed underground parallel or nearby to the existing transfer pipelines. This pipeline shall replace the existing 24 inch fuel oil line in kind and shall connect to existing fuel lines in ML-1 and ML-2 where existing 24 inch fuel oil line connects currently. The existing metering assembly near Pier 3 need not be replaced. The new pipe shall connect to the marine loading arms at Pier 3 where the 24 inch line connects currently. The new line shall replace all of the risers from the existing 24 inch fuel oil line on Pier 3 and shall connect to auxiliary and stripping pipelines near building 612 where the existing 24 inch line connects currently. The project will include impressed current cathodic protection to the pipeline and will require the appropriate isolation valves, equipment, coating systems for aboveground and underground piping, and accessories similar to what has been provided for the existing transfer pipelines. The project also provides high point vent and low point drain connection, thermal relief, and all other appurtenances required by UFC 3-460-01. Valves in the 20 inch line shall be full bore plug valves (dual block and bleed). Valves connecting the new pipeline to laterals in ML1 and ML-2 shall match the diameter of the lateral pipeline.

1.2.2 Location

The work is located at the NAVSTA Rota, Spain, approximately as indicated. The exact location will be shown by the Contracting Officer.

1.3 CONTRACT DRAWINGS

The following drawings accompany this specification and are a part thereof.

SHEET NUMBER / NÚMERO DE PLANO	NAVFAC NO.	SHEET TITLE	TÍTULO DE PLANO
GENERAL / GENERAL			
G-001	14142702	COVER SHEET	PORTADA
G-002	14142703	INDEX - SHEET 1	ÍNDICE - PLANO 1
G-003	14142704	INDEX - SHEET 2	ÍNDICE - PLANO 2
G-004	14142705	GENERAL ABBREVIATION SHEET	PLANO DE ABREVIATURAS GENERALES
CIVIL: CIVIL-GENERAL / URBANIZACIÓN: URBANIZACIÓN - GENERAL			
C-001	14142706	CIVIL LEGEND - SHEET 1	LEYENDA DE URBANIZACIÓN - PLANO 1
C-002	14142707	CIVIL LEGEND - SHEET 2	LEYENDA DE URBANIZACIÓN - PLANO 2
CIVIL: CIVIL DEMO / URBANIZACIÓN: URBANIZACIÓN - DEMOLICIONES			
CD101	14142708	SITE DEMOLITION KEY PLAN	PLANTA DE REFERENCIA DE DEMOLICIÓN DE EXTERIORES
CD106	14142709	DEMOLITION PLAN	PLANTA DE DEMOLICIÓN
CD107	14142710	DEMOLITION PLAN	PLANTA DE DEMOLICIÓN
CD108	14142711	DEMOLITION PLAN	PLANTA DE DEMOLICIÓN
CD109	14142712	DEMOLITION PLAN	PLANTA DE DEMOLICIÓN
CD110	14142713	DEMOLITION PLAN	PLANTA DE DEMOLICIÓN

CD111	14142714	DEMOLITION PLAN	PLANTA DE DEMOLICIÓN
CD112	14142715	DEMOLITION PLAN	PLANTA DE DEMOLICIÓN
CD113	14142716	DEMOLITION PLAN	PLANTA DE DEMOLICIÓN
CD114	14142717	DEMOLITION PLAN	PLANTA DE DEMOLICIÓN
CD115	14142718	DEMOLITION PLAN	PLANTA DE DEMOLICIÓN
CD116	14142719	DEMOLITION PLAN	PLANTA DE DEMOLICIÓN
CD117	14142720	DEMOLITION PLAN	PLANTA DE DEMOLICIÓN
CD118	14142721	DEMOLITION PLAN	PLANTA DE DEMOLICIÓN
CD119	14142722	DEMOLITION PLAN	PLANTA DE DEMOLICIÓN
CD120	14142723	DEMOLITION PLAN	PLANTA DE DEMOLICIÓN
CIVIL: CIVIL-SITE / URBANIZACIÓN: URBANIZACIÓN - EXTERIORES			
C-101	14142724	OVERALL SITE KEY PLAN	PLANTA DE REFERENCIA DE EXTERIORES GENERAL
C-106	14142725	SITE PLAN	PLANTA DE EXTERIORES
C-107	14142726	SITE PLAN	PLANTA DE EXTERIORES
C-108	14142727	SITE PLAN	PLANTA DE EXTERIORES
C-109	14142728	SITE PLAN	PLANTA DE EXTERIORES
C-110	14142729	SITE PLAN	PLANTA DE EXTERIORES
C-111	14142730	SITE PLAN	PLANTA DE EXTERIORES

C-112	14142731	SITE PLAN	PLANTA DE EXTERIORES
C-113	14142732	SITE PLAN	PLANTA DE EXTERIORES
C-114	14142733	SITE PLAN	PLANTA DE EXTERIORES
C-115	14142734	SITE PLAN	PLANTA DE EXTERIORES
C-116	14142735	SITE PLAN	PLANTA DE EXTERIORES
C-117	14142736	SITE PLAN	PLANTA DE EXTERIORES
C-118	14142737	SITE PLAN	PLANTA DE EXTERIORES
C-119	14142738	SITE PLAN	PLANTA DE EXTERIORES
C-120	14142739	SITE PLAN	PLANTA DE EXTERIORES
C-401	14142740	ENLARGED SITE PLAN - SHEET 1	PLANTA AMPLIADA DE EXTERIORES - PLANO 1
C-402	14142741	ENLARGED SITE PLAN - SHEET 2	PLANTA AMPLIADA DE EXTERIORES - PLANO 2
C-403	14142742	ENLARGED SITE PLAN - SHEET 3	PLANTA AMPLIADA DE EXTERIORES - PLANO 3
C-404	14142743	ENLARGED SITE PLAN - SHEET 4	PLANTA AMPLIADA DE EXTERIORES - PLANO 4
C-501	14142744	CIVIL DETAILS	DETALLES DE URBANIZACIÓN
C-502	14142745	CIVIL DETAILS	DETALLES DE URBANIZACIÓN
C-503	14142746	CIVIL DETAILS	DETALLES DE URBANIZACIÓN
C-504	14142747	DETAILS - FENCING	DETALLES - VALLA

C-505	14142748	DETAILS - FENCING	DETALLES - VALLA
CIVIL: CIVIL GRADING / URBANIZACIÓN: NIVELACIÓN DE URBANIZACIÓN			
CG401	14142749	ENLARGED GRADING PLAN - SHEET 1	PLANTA AMPLIADA DE NIVELACIÓN - PLANO 1
CG402	14142750	ENLARGED GRADING PLAN - SHEET 2	PLANTA AMPLIADA DE NIVELACIÓN - PLANO 2
STRUCTURAL / ESTRUCTURA			
S-021	14142751	STRUCTURAL NOTES - SHEET 1	NOTAS DE ESTRUCTURA - PLANO 1
S-022	14142752	STRUCTURAL NOTES - SHEET 2	NOTAS DE ESTRUCTURA - PLANO 2
S-521	14142753	TYPICAL REINFORCING DETAILS	DETALLES TÍPICOS DE ARMADO
S-522	14142754	TYPICAL FOUNDATION DETAILS	DETALLES TÍPICOS DE CIMENTACIÓN
S-523	14142755	MISCELLANEOUS DETAILS	DETALLES VARIOS
S-524	14142756	MISCELLANEOUS DETAILS	DETALLES VARIOS
S-530	14142757	TYPICAL PIPE SUPPORT DETAILS	DETALLES TÍPICOS DE SOPORTES DE TUBERÍAS
MECHANICAL: MECHANICAL-GENERAL / MECÁNICA: MECÁNICA - GENERAL			
M-001	14142758	MECHANICAL LEGEND	LEYENDA DE MECÁNICA
MECHANICAL: MECHANICAL - DEMOLITION / MECÁNICA: MECÁNICA - DEMOLICIONES			
MD101	14142759	MECHANICAL DEMOLITION KEY PLAN	PLANTA DE REFERENCIA DE DEMOLICIÓN

MD102	14142760	MECHANICAL DEMOLITION PLAN	PLANTA DE DEMOLICIONES DE MECÁNICA
MD103	14142761	MECHANICAL DEMOLITION PLAN	PLANTA DE DEMOLICIONES DE MECÁNICA
MD104	14142762	MECHANICAL DEMOLITION PLAN	PLANTA DE DEMOLICIONES DE MECÁNICA
MD105	14142763	MECHANICAL DEMOLITION PLAN	PLANTA DE DEMOLICIONES DE MECÁNICA
MD106	14142764	MECHANICAL DEMOLITION PLAN	PLANTA DE DEMOLICIONES DE MECÁNICA
MD107	14142765	MECHANICAL DEMOLITION PLAN	PLANTA DE DEMOLICIONES DE MECÁNICA
MD108	14142766	MECHANICAL DEMOLITION PLAN	PLANTA DE DEMOLICIONES DE MECÁNICA
MD109	14142767	MECHANICAL DEMOLITION PLAN	PLANTA DE DEMOLICIONES DE MECÁNICA
MD110	14142768	MECHANICAL DEMOLITION PLAN	PLANTA DE DEMOLICIONES DE MECÁNICA
MD111	14142769	MECHANICAL DEMOLITION PLAN	PLANTA DE DEMOLICIONES DE MECÁNICA
MD119	14142770	MECHANICAL DEMOLITION PLAN	PLANTA DE DEMOLICIONES DE MECÁNICA
MD120	14142771	MECHANICAL DEMOLITION PLAN ML-1 TO ML-2	PLANTA DE DEMOLICIONES DE MECÁNICA DE ML1 A ML-2
MD401	14142772	DEMOLITION PLAN VP-1 MANHOLE	PLANTA DE DEMOLICIÓN DE POZO VP-1
MD402	14142773	DEMOLITION PLAN VP-1.5 MANHOLE	PLANTA DE DEMOLICIÓN DE POZO VP-1.5

MD403	14142774	DEMOLITION PLAN BPH-VP-1 MANHOLE	PLANTA DE DEMOLICIÓN DE POZO BPH-VP-1
MD404	14142775	DEMOLITION PLAN VP-3 & 3.5 MANHOLE	PLANTA DE DEMOLICIÓN DE POZO VP-3 Y 3.5
MD410	14142776	DEMOLITION PLAN ML-1 MANHOLE	PLANTA DE DEMOLICIÓN DE POZO ML-1
MD411	14142777	DEMOLITION PLAN ML-2 MANHOLE	PLANTA DE DEMOLICIÓN DE POZO ML-2
MECHANICAL: MECHANICAL - SITE / MECÁNICA: MECÁNICA - EXTERIORES			
M-101	14142778	OVERALL MECHANICAL SITE & KEY PLAN	PLANTA GENERAL EXTERIOR Y DE REFERENCIA
M-102	14142779	MECHANICAL SITE PLAN	PLANTA EXTERIOR DE MECÁNICA
M-103	14142780	MECHANICAL SITE PLAN	PLANTA EXTERIOR DE MECÁNICA
M-104	14142781	MECHANICAL SITE PLAN	PLANTA EXTERIOR DE MECÁNICA
M-105	14142782	MECHANICAL SITE PLAN	PLANTA EXTERIOR DE MECÁNICA
M-106	14142783	MECHANICAL SITE PLAN	PLANTA EXTERIOR DE MECÁNICA
M-107	14142784	MECHANICAL SITE PLAN	PLANTA EXTERIOR DE MECÁNICA
M-108	14142785	MECHANICAL SITE PLAN	PLANTA EXTERIOR DE MECÁNICA
M-109	14142786	MECHANICAL SITE PLAN	PLANTA EXTERIOR DE MECÁNICA
M-110	14142787	MECHANICAL SITE PLAN	PLANTA EXTERIOR DE MECÁNICA
M-111	14142788	MECHANICAL SITE PLAN	PLANTA EXTERIOR DE MECÁNICA

M-112	14142789	MECHANICAL SITE PLAN	PLANTA EXTERIOR DE MECÁNICA
M-113	14142790	MECHANICAL SITE PLAN	PLANTA EXTERIOR DE MECÁNICA
M-114	14142791	MECHANICAL SITE PLAN	PLANTA EXTERIOR DE MECÁNICA
M-115	14142792	MECHANICAL SITE PLAN	PLANTA EXTERIOR DE MECÁNICA
M-116	14142793	MECHANICAL SITE PLAN	PLANTA EXTERIOR DE MECÁNICA
M-117	14142794	MECHANICAL SITE PLAN	PLANTA EXTERIOR DE MECÁNICA
M-118	14142795	MECHANICAL SITE PLAN	PLANTA EXTERIOR DE MECÁNICA
M-119	14142796	MECHANICAL SITE PLAN	PLANTA EXTERIOR DE MECÁNICA
M-120	14142797	MECHANICAL SITE PLAN	PLANTA EXTERIOR DE MECÁNICA
M-401	14142798	ENLARGED SITE PLAN VP-1 MANHOLE	PLANTA EXTERIOR AMPLIADA DEL POZO VP-1
M-402	14142799	ENLARGED SITE PLAN BPH-VP-1.5 MANHOLE	PLANTA EXTERIOR AMPLIADA DEL POZO BPH-VP-1.5
M-403	14142800	ENLARGED SITE PLAN - BPH-VP-1 MANHOLE	PLANTA EXTERIOR AMPLIADA DEL POZO BPH-VP-1
M-404	14142801	ENLARGED SITE PLAN BPH-VP- 3 & 3.5 MANHOLE	PLANTA EXTERIOR AMPLIADA DEL POZO BPH-VP-3 Y 3.5
M-410	14142802	ENLARGED SITE PLAN - ML1	PLANTA EXTERIOR AMPLIADA - ML1
M-411	14142803	ENLARGED SITE PLAN - ML2	PLANTA EXTERIOR AMPLIADA - ML2
M-501	14142804	MECHANICAL DETAILS - SHEET 1	DETALLES DE MECÁNICA - PLANO 1
M-502	14142805	MECHANICAL DETAILS - SHEET 2	DETALLES DE MECÁNICA - PLANO 2

M-503	14142806	MECHANICAL FUELING DETAILS	DETALLES DE COMBUSTIBLE MECÁNICO
M-504	14142807	MECHANICAL FUELING DETAILS	DETALLES DE COMBUSTIBLE MECÁNICO
M-505	14142808	TYPICAL PIPING PLAN	PLANTA TÍPICA DE TUBERÍAS
M-506	14142809	MECHANICAL SECTION	SECCIÓN MECÁNICA
M-601	14142810	ALIGNMENT & PIPE SCHEDULES - SHEET 1	CUADROS DE ALINEACIONES Y TUBERÍAS - PLANO 1
M-602	14142811	ALIGNMENT & PIPE SCHEDULES - SHEET 2	CUADROS DE ALINEACIONES Y TUBERÍAS - PLANO 2
M-611	14142812	PROCESS FLOW DIAGRAM - SHEET 1	DIAGRAMA DE FLUJO DEL PROCESO - PLANO 1
M-612	14142813	PROCESS FLOW DIAGRAM - SHEET 2	DIAGRAMA DE FLUJO DEL PROCESO - PLANO 2
M-613	14142814	PROCESS FLOW DIAGRAM - SHEET 3	DIAGRAMA DE FLUJO DEL PROCESO - PLANO 3
M-614	14142815	PROCESS FLOW DIAGRAM - SHEET 4	DIAGRAMA DE FLUJO DEL PROCESO - PLANO 4
M-615	14142816	PROCESS FLOW DIAGRAM - SHEET 5	DIAGRAMA DE FLUJO DEL PROCESO - PLANO 5
M-616	14142817	PROCESS FLOW DIAGRAM - SHEET 6	DIAGRAMA DE FLUJO DEL PROCESO - PLANO 6
ELECTRIC: ELECTRIC GENERAL / ELECTRICIDAD: ELECTRICIDAD GENERAL			
E-001	14142818	ELECTRICAL SYMBOLS & LEGEND SHEET	PLANO DE SÍMBOLOS ELÉCTRICOS Y LEYENDA
E-002	14142819	ELECTRICAL SYMBOLS & LEGEND SHEET	PLANO DE SÍMBOLOS ELÉCTRICOS Y LEYENDA

E-003	14142820	ELECTRICAL SYMBOLS & LEGEND SHEET	PLANO DE SÍMBOLOS ELÉCTRICOS Y LEYENDA
E-004	14142821	ELECTRICAL SYMBOLS & LEGEND SHEET	PLANO DE SÍMBOLOS ELÉCTRICOS Y LEYENDA
E-005	14142822	GENERAL NOTES SHEET	PLANO DE NOTAS GENERALES
E-006	14142823	GENERAL NOTES SHEET	PLANO DE NOTAS GENERALES
ELECTRIC: ELECTRIC DEMOLITION / ELECTRICIDAD: DEMOLICIONES ELÉCTRICAS			
ED401	14142824	ELECTRIC DEMOLITION PLAN - VP-1 MANHOLE	PLANTA DE DEMOLICIÓN ELÉCTRICA DE POZO VP-1
ED404	14142825	ELECTRIC DEMOLITION PLAN - VP-3 & 3.5 MANHOLE	PLANTA DE DEMOLICIÓN ELÉCTRICA DE POZO VP-3 Y 3.5
ELECTRIC: ELECTRIC SITE / ELECTRICIDAD: EXTERIOR DE ELECTRICIDAD			
E-401	14142826	ELECTRIC VAULT PLAN BPH-VP-1	PLANTA DE ARQUETA ELÉCTRICA BPH-VP-1
E-404	14142827	ELECTRIC VAULT BPH-VP-3 & 3.5	ARQUETA ELÉCTRICA BPH-VP-3 Y 3.5
E-501	14142828	ELECTRIC DETAILS	DETALLES DE ELECTRICIDAD
E-601	14142829	P3-1-VP1 CONTROL PANEL 1 LINE DIAGRAM	DIAGRAMA UNIFILAR DEL PANEL 1 DE CONTROL P3-1-VP1
E-607	14142830	BPH-VP-3 & 3.5 1-LINE DIAGRAM	DIAGRAMA UNIFILAR DEL PANEL DE CONTROL RÍO SALADO ESTE
ELECTRIC: CATHODIC PROTECTION / ELECTRICIDAD: PROTECCIÓN CATÓDICA			
EC101	14142831	CATHODIC PROTECTION KEY PLAN	PLANTA DE REFERENCIA DE PROTECCIÓN CATÓDICA
EC106	14142832	CATHODIC PROTECTION PLAN	PLANTA DE PROTECCIÓN CATÓDICA

EC107	14142833	CATHODIC PROTECTION PLAN	PLANTA DE PROTECCIÓN CATÓDICA
EC108	14142834	CATHODIC PROTECTION PLAN	PLANTA DE PROTECCIÓN CATÓDICA
EC109	14142835	CATHODIC PROTECTION PLAN	PLANTA DE PROTECCIÓN CATÓDICA
EC110	14142836	CATHODIC PROTECTION PLAN	PLANTA DE PROTECCIÓN CATÓDICA
EC111	14142837	CATHODIC PROTECTION PLAN	PLANTA DE PROTECCIÓN CATÓDICA
EC112	14142838	CATHODIC PROTECTION PLAN	PLANTA DE PROTECCIÓN CATÓDICA
EC113	14142839	CATHODIC PROTECTION PLAN	PLANTA DE PROTECCIÓN CATÓDICA
EC114	14142840	CATHODIC PROTECTION PLAN	PLANTA DE PROTECCIÓN CATÓDICA
EC115	14142841	CATHODIC PROTECTION PLAN	PLANTA DE PROTECCIÓN CATÓDICA
EC116	14142842	CATHODIC PROTECTION PLAN	PLANTA DE PROTECCIÓN CATÓDICA
EC117	14142843	CATHODIC PROTECTION PLAN	PLANTA DE PROTECCIÓN CATÓDICA
EC118	14142844	CATHODIC PROTECTION PLAN	PLANTA DE PROTECCIÓN CATÓDICA
EC119	14142845	CATHODIC PROTECTION PLAN	PLANTA DE PROTECCIÓN CATÓDICA
EC120	14142846	CATHODIC PROTECTION PLAN	PLANTA DE PROTECCIÓN CATÓDICA
EC501	14142847	CATHODIC PROTECTION DETAILS	DETALLES DE PROTECCIÓN CATÓDICA
ELECTRIC GROUNDING/			
EG101	14142848	ELECTRICAL GROUNDING SITE PLAN AND KEY MAP	PLANTA DE REFERENCIA DE EXTERIORES GENERAL

EG401	14142849	ELECTRICAL ENLARGED GROUNDING PLANS	PLANTAS ELÉCTRICAS AMPLIADAS DE PUESTA A TIERRA
EG501	14142850	ELECTRICAL GROUNDING DETAILS	DETALLES
REFERENCE SHEETS/ PLANOS DE REFERENCIA			
SURVEY / TOPOGRAFÍA			
V-001	N/A	NOTES, LEGEND AND SYMBOLS	NOTAS, LEYENDAS Y SIMBOLOS
V-002	N/A	BENCHMARKS	BASES DE REPLANTEO
V-003	N/A	BENCHMARKS	BASES DE REPLANTEO
V-004	N/A	BENCHMARK SCHEDULES	CUADROS DE BASES DE REPLANTEO
V-005	N/A	BENCHMARK SCHEDULES	CUADROS DE BASES DE REPLANTEO
V-006	N/A	BENCHMARK SCHEDULES	CUADROS DE BASES DE REPLANTEO
V-101	N/A	GENERAL SURVEY - EN -ROUTE SITE	PLANTA GENERAL DE TOPOGRAFIA - SITIO EN-ROUTE
V-102	N/A	GENERAL SURVEY - NEW PIPE	PLANTA GENERAL DE TOPOGRAFÍA - NUEVA TUBERÍA
V-401	N/A	ENLARGED SURVEY - AREA I - EN-ROUTE SITE	PLANTA AMPLIADA DE TOPOGRAFÍA - AREA I - SITIO EN-ROUTE
V-402	N/A	ENLARGED SURVEY - AREA II - EN-ROUTE SITE	PLANTA AMPLIADA DE TOPOGRAFÍA - AREA II - SITIO EN-ROUTE
V-403	N/A	ENLARGED SURVEY - AREA III - EN-ROUTE SITE	PLANTA AMPLIADA DE TOPOGRAFÍA - AREA III - SITIO EN-ROUTE
V-404	N/A	ENLARGED SURVEY - AREA IV - EN-ROUTE SITE	PLANTA AMPLIADA DE TOPOGRAFÍA - AREA IV - SITIO EN-ROUTE

V-405	N/A	ENLARGED SURVEY - AREA I - NEW PIPE	PLANTA AMPLIADA DE TOPOGRAFÍA - AREA I - NUEVA TUBERÍA
V-406	N/A	ENLARGED SURVEY - AREA II - NEW PIPE	PLANTA AMPLIADA DE TOPOGRAFÍA - AREA II - NUEVA TUBERÍA
V-407	N/A	ENLARGED SURVEY - AREA III - NEW PIPE	PLANTA AMPLIADA DE TOPOGRAFÍA - AREA III - NUEVA TUBERÍA
V-408	N/A	ENLARGED SURVEY - AREA IV - NEW PIPE	PLANTA AMPLIADA DE TOPOGRAFÍA - AREA IV - NUEVA TUBERÍA
V-409	N/A	ENLARGED SURVEY - AREA V - NEW PIPE	PLANTA AMPLIADA DE TOPOGRAFÍA - AREA V - NUEVA TUBERÍA
V-410	N/A	ENLARGED SURVEY - AREA VI - NEW PIPE	PLANTA AMPLIADA DE TOPOGRAFÍA - AREA VI - NUEVA TUBERÍA
V-411	N/A	ENLARGED SURVEY - AREA VII - NEW PIPE	PLANTA AMPLIADA DE TOPOGRAFÍA - AREA VII - NUEVA TUBERÍA
V-412	N/A	ENLARGED SURVEY - AREA VIII - NEW PIPE	PLANTA AMPLIADA DE TOPOGRAFÍA - AREA VIII - NUEVA TUBERÍA
V-413	N/A	ENLARGED SURVEY - AREA IX - NEW PIPE	PLANTA AMPLIADA DE TOPOGRAFÍA - AREA IX - NUEVA TUBERÍA
V-414	N/A	ENLARGED SURVEY - AREA X - NEW PIPE	PLANTA AMPLIADA DE TOPOGRAFÍA - AREA X - NUEVA TUBERÍA
V-415	N/A	ENLARGED SURVEY - AREA XI - NEW PIPE	PLANTA AMPLIADA DE TOPOGRAFÍA - AREA XI - NUEVA TUBERÍA
V-416	N/A	ENLARGED SURVEY - AREA XII - NEW PIPE	PLANTA AMPLIADA DE TOPOGRAFÍA - AREA XII - NUEVA TUBERÍA
V-417	N/A	ENLARGED SURVEY - AREA XIII - NEW PIPE	PLANTA AMPLIADA DE TOPOGRAFÍA - AREA XIII - NUEVA TUBERÍA
V-418	N/A	ENLARGED SURVEY - AREA XIV - NEW PIPE	PLANTA AMPLIADA DE TOPOGRAFÍA - AREA XIV - NUEVA TUBERÍA

V-419	N/A	ENLARGED SURVEY - AREA XV - NEW PIPE	PLANTA AMPLIADA DE TOPOGRAFÍA - AREA XV - NUEVA TUBERÍA
V-420	N/A	ENLARGED SURVEY - AREA XVI - NEW PIPE	PLANTA AMPLIADA DE TOPOGRAFÍA - AREA XVI - NUEVA TUBERÍA
V-421	N/A	ENLARGED SURVEY - AREA XVII - NEW PIPE	PLANTA AMPLIADA DE TOPOGRAFÍA - AREA XVII - NUEVA TUBERÍA
V-422	N/A	ENLARGED SURVEY - AREA XVIII - NEW PIPE	PLANTA AMPLIADA DE TOPOGRAFÍA - AREA XVIII - NUEVA TUBERÍA
V-423	N/A	ENLARGED SURVEY - AREA XIX - NEW PIPE	PLANTA AMPLIADA DE TOPOGRAFÍA - AREA XIX - NUEVA TUBERÍA
V-424	N/A	ENLARGED SURVEY - NEW PIPE	PLANTA AMPLIADA DE TOPOGRAFÍA - NUEVA TUBERÍA
V-425	N/A	ENLARGED SURVEY - DETAIL BPH-VP-1.5 MANHOLE	PLANTA AMPLIADA - DETALLE DE ARQUETA BPH-VP-1.5
V-426	N/A	ENLARGED SURVEY - DETAIL BPH-VP-2 MANHOLE	PLANTA AMPLIADA - DETALLE DE ARQUETA BPH-VP-2
V-427	N/A	ENLARGED SURVEY - DETAIL BPH-VP-3 MANHOLE	PLANTA AMPLIADA - DETALLE DE ARQUETA BPH-VP-3
V-428	N/A	ENLARGED SURVEY - DETAIL ESPAGUETI MANHOLE	PLANTA AMPLIADA - DETALLE DE ARQUETA ESPAGUETI
V-429	N/A	ENLARGED SURVEY - PIPING DETAILS SALADO RIVER CROSSING	PLANTA AMPLIADA - DETALLE PASO DE TUBERIAS RIO SALADO
V-430	N/A	ENLARGED SURVEY - AREA II - NEW PIPE	PLANTA AMPLIADA DE TOPOGRAFÍA - AREA II - NUEVA TUBERÍA
V-431	N/A	ENLARGED SURVEY - ML-1	PLANTA AMPLIADA DE TOPOGRAFÍA - ML-1

V-432	N/A	ENLARGED SURVEY - ML-2	PLANTA AMPLIADA DE TOPOGRAFÍA - ML-2
V-433	N/A	ENLARGED SURVEY - ML-3	PLANTA AMPLIADA DE TOPOGRAFÍA - ML-3
V-434	N/A	ENLARGED SURVEY - ML-4	PLANTA AMPLIADA DE TOPOGRAFÍA - ML-4
GEOTECHNICAL / GEOTECNIA			
B-001	N/A	GENERAL NOTES	NOTAS GENERALES
BJ101	N/A	BORINGS LOCATION	SITUACIÓN DE SONDEOS
BJ102	N/A	BORINGS LOCATION - KEY PLAN - NEW PIPE	SITUACIÓN DE SONDEOS - PLANTA DE REFERENCIA - NUEVA TUBERÍA
BJ103	N/A	BORINGS LOCATION - AREA I - NEW PIPE	SITUACIÓN DE SONDEOS - ÁREA I - NUEVA TUBERÍA
BJ104	N/A	BORINGS LOCATION - AREA II - NEW PIPE	SITUACIÓN DE SONDEOS - ÁREA II - NUEVA TUBERÍA
BJ105	N/A	BORINGS LOCATION - AREA III - NEW PIPE	SITUACIÓN DE SONDEOS - ÁREA III - NUEVA TUBERÍA
BJ106	N/A	BORINGS LOCATION - AREA IV - NEW PIPE	SITUACIÓN DE SONDEOS - ÁREA IV - NUEVA TUBERÍA
BJ107	N/A	BORINGS LOCATION - AREA V - NEW PIPE	SITUACIÓN DE SONDEOS - ÁREA V - NUEVA TUBERÍA
BJ108	N/A	BORINGS LOCATION - AREA VI - NEW PIPE	SITUACIÓN DE SONDEOS - ÁREA VI - NUEVA TUBERÍA
BJ301	N/A	PROFILES	PERFILES

BJ302	N/A	PROFILES	PERFILES
BJ601	N/A	BORINGS LOGOS - EN ROUTE SITE	REGISTROS DE SONDEOS - SITIO EN-ROUTE
BJ602	N/A	BORINGS LOGOS - EN ROUTE SITE	REGISTROS DE SONDEOS - SITIO EN-ROUTE
BJ603	N/A	BORINGS LOGOS - EN ROUTE SITE	REGISTROS DE SONDEOS - SITIO EN-ROUTE
BJ604	N/A	BORINGS LOGOS - EN ROUTE SITE	REGISTROS DE SONDEOS - SITIO EN-ROUTE
BJ605	N/A	BORING LOGOS - NEW PIPE SITE	REGISTROS DE SONDEOS - SITIO NUEVA TUBERÍA
BJ606	N/A	BORING LOGOS - NEW PIPE SITE	REGISTROS DE SONDEOS - SITIO NUEVA TUBERÍA
BJ607	N/A	BORING LOGOS - NEW PIPE SITE	REGISTROS DE SONDEOS - SITIO NUEVA TUBERÍA
BJ608	N/A	BORING LOGOS - NEW PIPE SITE	REGISTROS DE SONDEOS - SITIO NUEVA TUBERÍA
BJ609	N/A	BORING LOGOS - NEW PIPE SITE	REGISTROS DE SONDEOS - SITIO NUEVA TUBERÍA
BJ610	N/A	BORING LOGOS - NEW PIPE SITE	REGISTROS DE SONDEOS - SITIO NUEVA TUBERÍA
BJ611	N/A	BORING LOGOS - NEW PIPE SITE	REGISTROS DE SONDEOS - SITIO NUEVA TUBERÍA
BJ612	N/A	SCHEDULES	CUADROS
BJ613	N/A	SCHEDULES	CUADROS
BJ614	N/A	SCHEDULES	CUADROS

BJ615	N/A	SCHEDULES	CUADROS
ELECTRICAL / ELECTRICIDAD			
P8	N/A	POWER CABLE FLOW DIAGRAMS - PIER 3, METER RUN AND SHORE BLOCKING VALVES	DIAGRAMAS DE FLUJO DEL CABLE DE FUERZA - MUELLE 3, CONTADOR Y VALVULAS DE BLOQUEO
E3	N/A	ELECTRICAL INSTALLATION DETAILS - 480V PANEL SCHEDULE - NEW PANELS	DETALLES DE LA INSTALACIÓN ELÉCTRICA - CUADRO DEL PANEL DE 480V-NUEVOS PANELES
PD24	N/A	PIPING AND INSTRUMENTATION DIAGRAMS - PIER 3 VALVE PIT 1	DIAGRAMAS DE TUBERÍA INSTRUMENTACIÓN - FOSA DE LA VÁLVULA 1 MUELLE 3
T21	N/A	TRENCHING AND CONDUIT ROUTING - VALVE PIT P3-1 AND PIER 3 METER RUN AREA	TRAZADO DE ZANJA Y TENDIDO DE CONDUCTOS - FOSA DE LA VÁLVULA P3-1 Y ZONA DEL CONTADOR MUELLE 3
PD31	N/A	PIPING AND INSTRUMENTATION DIAGRAMS - BOOSTER PUMP HOUSE VALVE PIT PIER 3	DIAGRAMAS DE TUBERÍA INSTRUMENTACIÓN - FOSA DE LA VÁLVULA 1 MUELLE 3
P6	N/A	POWER CABLE FLOW DIAGRAMS - RIO SALADO EAST AND WEST VALVE PITS, VP 2 AND VALVE PIT PP	DIAGRAMAS DE FLUJO DEL CABLE DE FUERZA - - FOSAS DE LA VÁLVULA DE RÍO SALADO ESTE Y OESTE, VP2 Y FOSA VÁLVULA PP
E4	N/A	ELECTRICAL INSTALLATION DETAILS - 480V PANEL SCHEDULE - NEW PANELS	DETALLES DE LA INSTALACIÓN ELÉCTRICA - CUADRO DEL PANEL DE 480V - NUEVOS PLANELES
PD40	N/A	PIPING AND INSTRUMENTATION DIAGRAMS - BOSTER PUMP HOUSE RIO SALADO VALVE PIT	DIAGRAMAS DE TUBERÍA INSTRUMENTACIÓN - FOSAS DE VÁLVULA RÍO SALADO DEL CUARTO DE BOMBAS DE REFUERZO

T19	N/A	TRENCHING AND CONDUIT ROUTING - BOOSTER PUMP HOUSE TANKS 203 THRU 210 AND 212	TRAZADO DE ZANJA Y TENDIDO DE CONDUCTOS - TANQUES DEL CUARTO DE BOMBAS DE REFUERZO DE 203 A 210 Y 212
-----	-----	---	--

Five sets of full size contract drawings, maps, and specifications will be furnished to the Contractor without charge. Reference publications will not be furnished. Immediately check furnished drawings and notify the Government of any discrepancies.

1.4 LOCATION OF UNDERGROUND UTILITIES

Obtain digging permits prior to start of excavation, and comply with Installation requirements for locating and marking underground utilities. Contact local utility locating service a minimum of 48 hours prior to excavating, to mark utilities, and within sufficient time required if work occurs on a Monday or after a Holiday. Verify existing utility locations indicated on contract drawings, within area of work.

Identify and mark all other utilities not managed and located by the local utility companies. Scan the construction site with Ground Penetrating Radar (GPR), electromagnetic, or sonic equipment, and mark the surface of the ground or paved surface where existing underground utilities or

utilities encased in pier structures are discovered. Verify the elevations of existing piping, utilities, and any type of underground or encased obstruction not indicated, or specified to be removed, that is indicated or discovered during scanning, in locations to be traversed by piping, ducts, and other work to be conducted or installed. Verify elevations before installing new work closer than nearest manhole or other structure at which an adjustment in grade can be made.

1.4.1 Notification Prior to Excavation

Notify the Contracting Officer at least 48 hours prior to starting excavation work.

1.5 SALVAGE MATERIAL AND EQUIPMENT

Items designated by the Contracting Officer to be salvaged remain the property of the Government. Segregate, itemize, deliver and off-load the salvaged property at the Government designated storage area located within 5.0 kilometers of the construction site.

Provide a salvage plan, listing material and equipment to be salvaged, and their storage location. Maintain property control records for material or equipment designated as salvage. Provide a system for property control in the salvage plan. Store and protect salvaged materials and equipment until disposition by the Contracting Officer.

PART 2 PRODUCTS

Not used.

PART 3 EXECUTION

Not used.

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 02 - EXISTING CONDITIONS

SECTION 02 41 00

DEMOLITION

08/22

PART 1 GENERAL

- 1.1 REFERENCES
- 1.2 PROJECT DESCRIPTION
 - 1.2.1 Definitions
 - 1.2.1.1 Demolition
 - 1.2.1.2 Demolition Plan
 - 1.2.2 Demolition Plan
 - 1.2.3 General Requirements
- 1.3 ITEMS TO REMAIN IN PLACE
 - 1.3.1 Existing Construction Limits and Protection
 - 1.3.2 Trees
 - 1.3.3 Utility Service
 - 1.3.4 Facilities
- 1.4 BURNING
- 1.5 AVAILABILITY OF WORK AREAS
- 1.6 SUBMITTALS
- 1.7 QUALITY ASSURANCE
 - 1.7.1 Dust and Debris Control
- 1.8 PROTECTION
 - 1.8.1 Traffic Control Signs
 - 1.8.2 Protection of Personnel
- 1.9 RELOCATIONS
- 1.10 EXISTING CONDITIONS

PART 2 PRODUCTS

- 2.1 FILL MATERIAL

PART 3 EXECUTION

- 3.1 EXISTING FACILITIES TO BE REMOVED
 - 3.1.1 Structures
 - 3.1.2 Utilities and Related Equipment
 - 3.1.2.1 General Requirements
 - 3.1.2.2 Disconnecting Existing Utilities
 - 3.1.3 Chain Link Fencing
 - 3.1.4 Paving and Slabs
 - 3.1.5 Concrete
 - 3.1.6 Structural Steel
 - 3.1.7 Miscellaneous Metal
 - 3.1.8 Patching
 - 3.1.9 Mechanical Equipment and Fixtures
 - 3.1.9.1 Piping
 - 3.1.9.2 Fuel Piping
 - 3.1.10 Electrical Equipment and Fixtures

- 3.1.10.1 Fixtures
- 3.1.10.2 Electrical Devices
- 3.1.10.3 Wiring Ducts or Troughs
- 3.1.10.4 Conduit and Miscellaneous Items
- 3.1.11 Items With Unique/Regulated Disposal Requirements
- 3.2 CONCURRENT EARTH-MOVING OPERATIONS
- 3.3 DISPOSITION OF MATERIAL
 - 3.3.1 Title to Materials
 - 3.3.2 Reuse of Materials and Equipment
 - 3.3.3 Salvaged Materials and Equipment
 - 3.3.4 Unsalvageable and Non-Recyclable Material
- 3.4 CLEANUP
- 3.5 DISPOSAL OF REMOVED MATERIALS
 - 3.5.1 Regulation of Removed Materials
 - 3.5.2 Burning on Government Property
 - 3.5.3 Removal from Government Property
- 3.6 REUSE OF SALVAGED ITEMS

-- End of Section Table of Contents --

SECTION 02 41 00

DEMOLITION
08/22

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN SOCIETY OF SAFETY PROFESSIONALS (ASSP)

ASSP A10.6 (2006) Safety & Health Program
Requirements for Demolition Operations -
American National Standard for
Construction and Demolition Operations

U.S. ARMY CORPS OF ENGINEERS (USACE)

EM 385-1-1 (2014) Safety -- Safety and Health
Requirements Manual

U.S. DEPARTMENT OF DEFENSE (DOD)

UFC 3-460-01 (2019; with Change 3, 2023) Design:
Petroleum Fuel Facilities

JUNTA DE ANDALUCIA (JA)

Decreto 73/2012 (2012) Decreto 73/2012, de 22 de marzo,
por el que se aprueba el Reglamento de
Residuos de Andalucía.

MINISTERIO DE MEDIO AMBIENTE (MA)

Orden MAM/304/2002 (2002) Orden MAM/304/2002, de 8 de
febrero, por la que se publican las
operaciones de valorización y eliminación
de residuos y la lista europea de residuos.

U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)

40 CFR 61 National Emission Standards for Hazardous
Air Pollutants

1.2 PROJECT DESCRIPTION

1.2.1 Definitions

1.2.1.1 Demolition

Demolition is the process of tearing apart and removing any feature of a facility together with any related handling and disposal operations.

1.2.1.2 Demolition Plan

Demolition Plan is the planned steps and processes for managing demolition activities and identifying the required sequencing activities and disposal mechanisms.

1.2.2 Demolition Plan

Prepare a [Demolition Plan](#) and submit proposed salvage, demolition, and removal procedures for approval before work is started. Include in the plan procedures for careful removal and disposition of materials specified to be salvaged, coordination with other work in progress, a disconnection schedule of utility services, and a detailed description of methods and equipment to be used for each operation and of the sequence of operations. [Comply with the requirements and methods of Decreto 73/2012 and Orden MAM/304/2002.](#) Identify components and materials to be salvaged for reuse or recycling with reference to paragraph Existing Facilities to be Removed. Append tracking forms for all removed materials indicating type, quantities, condition, destination, and end use. Coordinate with Waste Management Plan in accordance with Section [01 74 19 CONSTRUCTION WASTE MANAGEMENT AND DISPOSAL](#). Provide procedures for safe conduct of the work in accordance with [EM 385-1-1](#). Plan must be approved by Contracting Officer prior to work beginning.

1.2.3 General Requirements

Do not begin demolition until authorization is received from the Contracting Officer. [The work of this section is to be performed in a manner that maximizes the value derived from the salvage and recycling of materials.](#) The work includes demolition, salvage of identified items and materials, and removal of resulting rubbish and debris. Remove rubbish and debris from Government property daily, unless otherwise directed. Store materials that cannot be removed daily in areas specified by the Contracting Officer. In the interest of occupational safety and health, perform the work in accordance with [EM 385-1-1](#), Section 23, Demolition, and other applicable Sections.

1.3 ITEMS TO REMAIN IN PLACE

Comply with FAR 52.236-9 to protect existing vegetation, structures, equipment, utilities, and improvements. Coordinate the work of this section with all other work indicated. Construct and maintain shoring, bracing, and supports as required. Ensure that structural elements are not overloaded. Increase structural supports or add new supports as may be required as a result of any cutting, removal, or demolition work performed under this contract. Do not overload structural elements [or](#) pavements to remain. Provide new supports and reinforcement for existing construction weakened by demolition, or removal work. Repairs, reinforcement, or structural replacement require approval by the Contracting Officer prior to performing such work.

1.3.1 Existing Construction Limits and Protection

Do not disturb existing construction beyond the extent indicated or necessary for installation of new construction. Provide temporary shoring and bracing for support of building components to prevent settlement or other movement. Provide protective measures to control accumulation and migration of dust and dirt in all work areas. Remove dust, dirt, and debris from work areas daily.

1.3.2 Trees

Protect trees within the project site which might be damaged during demolition, and which are indicated to be left in place, by a 1.8 m high fence. Erect and secure fence a minimum of 1.5 m from the trunk of individual trees or follow the outer perimeter of branches or clumps of trees. Replace any tree designated to remain that is damaged during the work under this contract with like-kind or as approved by the Contracting Officer.

1.3.3 Utility Service

Maintain existing utilities indicated to stay in service and protect against damage during demolition operations. Prior to start of work, utilities serving each area of alteration or removal will be shut off by the Government and disconnected and sealed by the Contractor.

1.3.4 Facilities

Protect electrical and mechanical services and utilities. Where removal of existing utilities and pavement is specified or indicated, provide approved barricades, temporary covering of exposed areas, and temporary services or connections for electrical and mechanical utilities. Floors, roofs, walls, columns, pilasters, and other structural components that are designed and constructed to stand without lateral support or shoring, and are determined to be in stable condition, must remain standing without additional bracing, shoring, or lateral support until demolished, unless directed otherwise by the Contracting Officer. Ensure that no elements determined to be unstable are left unsupported and place and secure bracing, shoring, or lateral supports as may be required as a result of any cutting, removal, or demolition work performed under this contract.

1.4 BURNING

The use of burning at the project site for the disposal of refuse and debris will not be permitted.

1.5 AVAILABILITY OF WORK AREAS

Areas in which the work is to be accomplished will be available at contract award.

1.6 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Demolition Plan; G

Existing Conditions

1.7 QUALITY ASSURANCE

Notify the Contracting Officer in writing 10 working days prior to the

commencement of work in accordance with 40 CFR 61, Subpart M. Comply with local and regional hauling and disposal regulations. In addition to the requirements of the "Contract Clauses," conform to the safety requirements contained in ASSP A10.6. Comply with the Environmental requirements specified. Use of explosives will not be permitted.

1.7.1 Dust and Debris Control

Prevent the spread of dust and debris and avoid the creation of a nuisance in the surrounding area. Do not use water if it results in hazardous or objectionable conditions such as, but not limited to, ice, flooding, or pollution.

1.8 PROTECTION

1.8.1 Traffic Control Signs

a. Where pedestrian and driver safety is endangered in the area of removal work, use traffic barricades with flashing lights. Anchor barricades in a manner to prevent displacement by wind. Notify the Contracting Officer prior to beginning such work.

1.8.2 Protection of Personnel

Before, during and after the demolition work continuously evaluate the condition of the site specific features being demolished and take immediate action to protect all personnel working in and around the project site. No area, section, or component of floors, roofs, walls, columns, pilasters, piping or other structural element will be allowed to be left standing without sufficient bracing, shoring, or lateral support to prevent collapse or failure while workmen remove debris or perform other work in the immediate area.

1.9 RELOCATIONS

Perform the removal and reinstallation of relocated items as indicated with workmen skilled in the trades involved. Repair or replace items to be relocated which are damaged by the Contractor with new undamaged items as approved by the Contracting Officer.

1.10 EXISTING CONDITIONS

Before beginning any demolition work, survey the site and examine the drawings and specifications to determine the extent of the work. Record existing conditions in the presence of the Contracting Officer showing the condition of structures and other facilities adjacent to areas of alteration or removal. Photographs or electronic images with a minimum resolution of 3072 x 2304 pixels, capable of a print resolution of 300 dpi, will be acceptable as a record of existing conditions. Include in the record the elevation of the top of foundations, piping and, possible conflicting electrical conduits, piping, etc. Include the location and extent of existing cracks and other damage and description of surface conditions that exist prior to starting work. It is the Contractor's responsibility to verify and document all required outages which will be required during the course of work, and to note these outages on the record document. Submit survey results to the Contracting Officer.

PART 2 PRODUCTS

2.1 FILL MATERIAL

- a. Comply with excavating, backfilling, and compacting procedures for soils used as backfill material to fill voids, depressions or excavations resulting from demolition of piping and structures. Provide fill material consisting of waste products from demolition until all waste appropriate for this purpose is consumed.
- b. Proposed fill material must be sampled and tested by an approved soil testing laboratory, as indicated in Section 31 23 00.00 20 EXCAVATION AND FILL:

PART 3 EXECUTION

3.1 EXISTING FACILITIES TO BE REMOVED

Dismantled and removed materials are to be separated, set aside, and prepared as specified, and stored or delivered to a collection point for reuse, remanufacture, recycling, or other disposal, as specified. Designate materials for reuse onsite whenever possible.

3.1.1 Structures

- a. Locate demolition equipment throughout the structure and remove materials so as to not impose excessive loads to supporting walls, floors, or framing.

3.1.2 Utilities and Related Equipment

3.1.2.1 General Requirements

Do not interrupt existing utilities serving occupied or used facilities, except when authorized in writing by the Contracting Officer. Do not interrupt existing utilities serving facilities occupied and used by the Government except when approved in writing and then only after temporary utility services have been approved and provided. Do not begin demolition work until all utility disconnections have been made. Shut off and cap utilities for future use, as indicated.

3.1.2.2 Disconnecting Existing Utilities

Remove existing utilities, as indicated and terminate in a manner conforming to the nationally recognized code covering the specific utility and approved by the Contracting Officer. When utility lines are encountered but are not indicated on the drawings, notify the Contracting Officer prior to further work in that area. Remove meters and related equipment and deliver to a location in accordance with instructions of the Contracting Officer.

3.1.3 Chain Link Fencing

Remove chain link fencing, gates and other related salvaged items scheduled for removal and transport to designated areas. Remove gates as whole units. Cut chain link fabric to 6 m lengths and store in rolls off the ground.

3.1.4 Paving and Slabs

Sawcut and remove concrete and asphaltic concrete paving and slabs including aggregate base as indicated. Provide neat sawcuts at limits of pavement removal as indicated. Move, grind and store pavement and slabs designated to be recycled and utilized in this project as directed by the Contracting Officer. Remove pavement and slabs not to be used in this project from the project site at Contractor's expense. If pavement remnant is 1 m or less, then remove and restore pavement.

3.1.5 Concrete

Saw concrete along straight lines to a depth of a minimum 50 mm. Make each cut in walls perpendicular to the face and in alignment with the cut in the opposite face. Break out the remainder of the concrete provided that the broken area is concealed in the finished work, and the remaining concrete is sound. At locations where the broken face cannot be concealed, grind smooth or saw cut entirely through the concrete. Sawcuts in existing concrete sidewalk, and roads and parking areas must be at the nearest existing expansion joint or weakened plane joint and at full depth.

3.1.6 Structural Steel

Dismantle structural steel at field connections and in a manner that will prevent bending or damage. Salvage for recycle structural steel, steel joists, girders, angles, plates, columns and shapes. Do not use flame-cutting torches. Transport steel joists and girders as whole units and not dismantled. Transport structural steel shapes to a designated recycling facility in the area directed by the Contracting Officer (close to Tow Way Road) and stored off the ground, protected from the weather.

3.1.7 Miscellaneous Metal

Salvage shop-fabricated items such as access doors and frames, steel gratings, metal ladders, wire mesh partitions, metal railings, metal windows and similar items as whole units. Salvage light-gage and cold-formed metal framing, such as steel studs, metal gutters and similar items. Provide separate containers to collect scrap metal and transport to a scrap metal collection or recycling facility close to Tow Way Road, in accordance with the Waste Management Plan.

3.1.8 Patching

Where removals leave holes and damaged surfaces exposed in the finished work, patch and repair these holes and damaged surfaces to match adjacent finished surfaces, using on-site materials when available. Where new work is to be applied to existing surfaces, perform removals and patching in a manner to produce surfaces suitable for receiving new work. Make finished surfaces of patched area flush with the adjacent existing surface and match the existing adjacent surface as closely as possible to texture and finish. Provide patching as specified and indicated, and include the following:

- a. Concrete and Masonry: Completely fill holes and depressions, caused by previous physical damage or left as a result of removals with an approved masonry patching material, applied in accordance with the manufacturer's printed instructions.

3.1.9 Mechanical Equipment and Fixtures

3.1.9.1 Piping

Disconnect piping at unions, flanges and valves, and fittings as required to reduce the pipe into straight lengths for practical storage. Store salvaged piping according to size and type. If the piping that remains can become pressurized due to upstream valve failure, attach end caps, blind flanges, or other types of plugs or fittings with a pressure gage and bleed valve to the open end of the pipe to ensure positive leak control. Carefully dismantle piping that previously contained gas, gasoline, oil, or other dangerous fluids, with precautions taken to prevent injury to persons and property. Store piping outdoors until all fumes and residues are removed. Box prefabricated supports, hangers, plates, valves, and specialty items according to size and type. Wrap sprinkler heads individually in plastic bags before boxing.

3.1.9.2 Fuel Piping

Drain, vacuum extract, pig (where possible), and purge fuel piping to be removed or abandoned in place in accordance with Chapter 14 of UFC 3-460-01

3.1.10 Electrical Equipment and Fixtures

Salvage motors, motor controllers, and operating and control equipment that are attached to the driven equipment. Salvage wiring systems and components. Box loose items and tag for identification. Disconnect primary, secondary, control, communication, and signal circuits at the point of attachment to their distribution system.

3.1.10.1 Fixtures

Remove and salvage electrical fixtures. Salvage unprotected glassware from the fixture and salvage separately. Salvage incandescent, mercury-vapor, and fluorescent lamps and fluorescent ballasts manufactured prior to 1978, boxed and tagged for identification, and protected from breakage.

3.1.10.2 Electrical Devices

Remove and salvage switches, switchgear, transformers, conductors including wire and nonmetallic sheathed and flexible armored cable, regulators, meters, instruments, plates, circuit breakers, panelboards, outlet boxes, and similar items. Box and tag these items for identification according to type and size.

3.1.10.3 Wiring Ducts or Troughs

Remove and salvage wiring ducts or troughs. Dismantle plug-in ducts and wiring troughs into unit lengths. Remove plug-in or disconnecting devices from the busway and store separately.

3.1.10.4 Conduit and Miscellaneous Items

Salvage conduit except where embedded in concrete or masonry. Consider corroded, bent, or damaged conduit as scrap metal. Sort straight and undamaged lengths of conduit according to size and type. Classify supports, knobs, tubes, cleats, and straps as debris to be removed and disposed.

3.1.11 Items With Unique/Regulated Disposal Requirements

Remove and dispose of items with unique or regulated disposal requirements in the manner dictated by law or in the most environmentally responsible manner.

3.2 CONCURRENT EARTH-MOVING OPERATIONS

Do not begin excavation, filling, and other earth-moving operations that are sequential to demolition work in areas occupied by structures to be demolished or deconstructed until all demolition in the area has been completed and debris removed. Fill holes, and other hazardous openings.

3.3 DISPOSITION OF MATERIAL

3.3.1 Title to Materials

Except for salvaged items, all materials and equipment removed and not reused or salvaged, become the property of the Contractor and must be removed from Government property. Materials approved for storage by the Contracting Officer must be removed before completion of the contract. Title to materials resulting from demolition, and materials and equipment to be removed, is vested in the Contractor upon approval by the Contracting Officer. The Government will not be responsible for the condition or loss of, or damage to, such property after contract award. Showing for sale or selling materials and equipment on site is prohibited.

3.3.2 Reuse of Materials and Equipment

Remove and store materials and equipment to be reused or relocated to prevent damage, and reinstall as the work progresses. Coordinate the re-use of materials and equipment with the re-use requirements in accordance with Section 01 74 19 CONSTRUCTION WASTE MANAGEMENT AND DISPOSAL. Capture re-use of materials in the diversion calculations for the project.

3.3.3 Salvaged Materials and Equipment

Remove materials and equipment that are indicated to be salvaged or are to be removed by the Contractor and that are to remain the property of the Government, and deliver to a storage site close to Tow Way Road.

- a. Salvage items and material to the maximum extent possible.
- b. Store all materials salvaged for the Contractor as approved by the Contracting Officer and remove from Government property before completion of the contract. Coordinate the salvaged materials with tracking requirements in accordance with Section 01 74 19 CONSTRUCTION WASTE MANAGEMENT AND DISPOSAL. Capture salvaged materials in the diversion calculations for the project.
- c. Remove salvaged items to remain the property of the Government in a manner to prevent damage, and packed or crated to protect the items from damage while in storage or during shipment. Items damaged during removal or storage must be repaired or replaced to match existing items. Properly identify the contents of containers.
- d. Remove the following items reserved as property of the using service

prior to commencement of work under this contract: signs, concrete barriers, vinyl fencing, guardrail, chain link fence swing gate, and bus stop shelter.

3.3.4 Unsalvageable and Non-Recyclable Material

Dispose of unsalvageable and non-recyclable noncombustible materials per Section 01 57 19.01 21 ENVIRONMENTAL CONTROLS. Dispose of unsalvageable and non-recyclable combustible material.

3.4 CLEANUP

Remove debris and rubbish from project site. Remove and transport the debris in a manner that prevents spillage on streets or adjacent areas. Apply local regulations regarding hauling and disposal.

3.5 DISPOSAL OF REMOVED MATERIALS

3.5.1 Regulation of Removed Materials

Dispose of debris, rubbish, scrap, and other nonsalvageable materials resulting from removal operations with all applicable local and regional regulations as contractually specified.

3.5.2 Burning on Government Property

Burning of materials removed from demolished and deconstructed structures will not be permitted on Government property.

3.5.3 Removal from Government Property

Transport waste materials removed from demolished structures, except waste soil, from Government property for legal disposal. Dispose of waste soil per Section 01 57 19.01 21 ENVIRONMENTAL CONTROLS.

3.6 REUSE OF SALVAGED ITEMS

Recondition salvaged materials and equipment designated for reuse before installation. Replace items damaged during removal and salvage operations or restore them as necessary to usable condition.

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 33 - UTILITIES

SECTION 33 52 40

FUEL SYSTEMS PIPING (NON-HYDRANT)

11/18, CHG 2: 11/20

PART 1 GENERAL

- 1.1 SUMMARY
- 1.2 REFERENCES
- 1.3 ADMINISTRATIVE REQUIREMENTS
- 1.4 SUBMITTALS
- 1.5 QUALITY ASSURANCE
 - 1.5.1 Contractor Qualifications
 - 1.5.2 Work Plan
 - 1.5.3 Pigging Plan
 - 1.5.4 Hydrotesting Plan
 - 1.5.5 Design Data
 - 1.5.5.1 Pipeline Inventory
 - 1.5.6 Delivery, Storage, and Handling

PART 2 PRODUCTS

- 2.1 ELECTRICAL WORK
 - 2.1.1 General
 - 2.1.2 Grounding and Bonding
- 2.2 MATERIALS
 - 2.2.1 Types of Fuel
 - 2.2.2 Carbon Steel Piping
 - 2.2.3 Stainless Steel Piping
 - 2.2.4 External Protective Coatings for Aboveground Piping
 - 2.2.5 External Protective Coatings for Buried Steel Piping
 - 2.2.5.1 Carbon Steel Piping
 - 2.2.6 Fittings
 - 2.2.6.1 General
 - 2.2.6.2 Carbon Steel Fittings
 - 2.2.6.3 Stainless Steel Fittings
 - 2.2.7 Insulating Flange Kits (Electrically Isolating)
 - 2.2.8 Bolts, Nuts and Washers
 - 2.2.9 Flange Gaskets, Non-Metallic, Non-Electrically Isolating
 - 2.2.9.1 Nitrile Butadiene (Buna-N)
 - 2.2.9.2 Acrylonitrile Butadiene Rubber (NBR)
 - 2.2.9.3 Polytetrafluoroethylene (PTFE)
 - 2.2.9.4 Fluoro Rubber FKM
 - 2.2.9.5 Fluoroelastomer FPM
- 2.3 MANUAL VALVES
 - 2.3.1 Ball Valves
 - 2.3.1.1 Materials
 - 2.3.2 Plug (Double Block and Bleed) Valves
 - 2.3.2.1 General
 - 2.3.2.2 Valve Operation
 - 2.3.2.3 Relief Valves

- 2.3.2.4 Bleed Valves
- 2.3.2.5 Electric Valve Actuator
- 2.4 THERMAL RELIEF VALVE
 - 2.4.1 Valve Material
 - 2.4.2 Thermal Relief Valve (ASME Type)
 - 2.4.3 Thermal Relief Valve (Balanced Type)
- 2.5 PIPING ACCESSORIES
 - 2.5.1 Mechanically Adjustable Segmented Elastomeric Seal
 - 2.5.2 Pipe Sleeves
 - 2.5.3 Pressure Gauge
 - 2.5.4 Differential Pressure Gauge
 - 2.5.5 Pipe Supports
 - 2.5.5.1 General
 - 2.5.5.2 Adjustable Pipe Supports
 - 2.5.5.3 Low Friction Supports
 - 2.5.5.4 U-bolt Half Round Supports
 - 2.5.5.5 Concrete and Grout
 - 2.5.6 Sample Connections
- 2.6 PIGGING SYSTEM COMPONENTS
 - 2.6.1 Maintenance Pig Launchers and Receivers
 - 2.6.2 Smart Pig Launchers and Receivers
 - 2.6.3 Launcher and Receiver Closure Door
 - 2.6.4 Signaler
- 2.7 MISCELLANEOUS ACCESSORIES
 - 2.7.1 Concrete Anchor Bolts
 - 2.7.2 Coatings for Bolts, Studs, Nuts, and Washers
 - 2.7.3 Polytetrafluoroethylene (PTFE) Tape
 - 2.7.4 Pipe Sleeves
 - 2.7.5 Buried Utility Tape
 - 2.7.6 Pipeline Markers
- 2.8 FINISHES
 - 2.8.1 Factory Coating
 - 2.8.1.1 Valves
 - 2.8.1.2 Equipment and Components
 - 2.8.2 Field Painting

PART 3 EXECUTION

- 3.1 GENERAL
- 3.2 VERIFICATION OF DIMENSIONS
- 3.3 CLEANING OF PIPING
- 3.4 TRENCHING AND BACKFILLING
- 3.5 PIPING LAYOUT REQUIREMENTS
 - 3.5.1 Pipe Fabrication
 - 3.5.2 Interferences and Measurements
 - 3.5.3 Space and Access
 - 3.5.4 Location
 - 3.5.5 Pipe Supports
 - 3.5.6 Structural Support
 - 3.5.7 Grade
 - 3.5.8 Size Changes
 - 3.5.9 Direction Changes
 - 3.5.10 Threaded End Connections
 - 3.5.11 Existing Pipe Systems
 - 3.5.12 Bolted Connections
 - 3.5.13 Flanges and Unions
 - 3.5.14 Manual Valves
 - 3.5.15 Air Vents
 - 3.5.16 Drains

- 3.5.17 Pipe Sleeves
- 3.6 STRUCTURAL ATTACHMENTS
- 3.7 WELDING
 - 3.7.1 General
- 3.8 INSTALLATION
 - 3.8.1 Precautions
 - 3.8.2 Protective Coatings for Buried Piping Including Stainless Steel Piping
 - 3.8.2.1 Application of Coating System
 - 3.8.2.2 Inspection and Testing
 - 3.8.2.3 Damage Repair
- 3.9 INTERIOR EPOXY COATING
- 3.10 INSTALLATION OF UNDERGROUND PIPE
 - 3.10.1 Pipe Assembly
 - 3.10.2 Warning Tapes in Earth Trenches
 - 3.10.3 Clearances
 - 3.10.4 Protective Coating
 - 3.10.5 Pipeline Markers
- 3.11 SYSTEM COMMISSIONING
- 3.12 TESTING
 - 3.12.1 Before Backfilling
 - 3.12.1.1 Exterior Coating Holiday Test
 - 3.12.2 Pneumatic Test
 - 3.12.2.1 Pneumatic Test Procedure
 - 3.12.3 Hydrostatic Test
 - 3.12.4 Soak Testing
 - 3.12.5 Performance Testing
- 3.13 PIPE PIGGING - CLEANING
 - 3.13.1 General
 - 3.13.2 Use of Fuel in Cleaning Pigging
 - 3.13.3 Cleaning Pig Run
 - 3.13.4 Wire Brush Pig Run
- 3.14 PIPE PIGGING VERIFICATION
 - 3.14.1 Geometry/Ultrasonic Tool Reports
 - 3.14.2 Pipeline Internal Inspection Operations
 - 3.14.2.1 General
 - 3.14.2.2 Preparatory Work
 - 3.14.2.3 Pig Load And Launch
 - 3.14.2.4 Pipeline Operation During Pigging
 - 3.14.2.5 Brush and Gauging Survey
 - 3.14.2.6 Geometry/Ultrasonic Survey
 - 3.14.2.7 Pipe Wall Thickness Survey
 - 3.14.2.8 Lost Pig

-- End of Section Table of Contents --

SECTION 33 52 40

FUEL SYSTEMS PIPING (NON-HYDRANT)
11/18, CHG 2: 11/20

PART 1 GENERAL

1.1 SUMMARY

This section defines the requirements for pipe, piping components, and valves as related to an non-hydrant, non-service station, fuel distribution system. Such systems include, but are not limited to: marine receipt, pipeline receipt, truck off-loading receipt, pump house, pump pad, truck loading, marine loading, transfer pipeline, product recovery, and other miscellaneous piping systems. Provide the entire fuel distribution system as a complete and fully operational system. Size, select, construct, and install equipment and system components to operate together as a complete system. Substitutions of functions specified herein will not be acceptable. Coordinate the work of the system manufacturer's service personnel during construction, testing, calibration, and acceptance of the system. System components and piping specified herein must be designed to handle a working pressure of 1900 kPa at 38 deg C. Components specified herein must be compatible with the fuel to be handled. Components to be suitable for outside, unsheltered location, and to function normally in ambient temperatures.

1.2 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN PETROLEUM INSTITUTE (API)

API RP 540	(1999; R 2004) Electrical Installations in Petroleum Processing Plants
API RP 1110	(2013; R 2018) Recommended Practice for the Pressure Testing of Steel Pipelines for the Transportation of Gas, Petroleum Gas, Hazardous Liquids, Highly Volatile Liquids, or Carbon Dioxide
API RP 1595	(2012; R 2019; 2nd Ed) Design, Construction, Operation, Maintenance, and Inspection of Aviation Pre-Airfield Storage Terminals
API RP 2003	(2015; 8th Ed) Protection Against Ignitions Arising out of Static, Lightning, and Stray Currents
API RP 2009	(2002; R 2007; 7th Ed) Safe Welding, Cutting, and Hot Work Practices in Refineries, Gasoline Plants, and Petrochemical Plants

API RP 2200	(1999; R 2004) Electrical Installations in Petroleum Processing Plants
API STD 608	(2012) Metal Ball Valves - Flanged, Threaded, And Welding End
API Spec 5L	(2018; 46th Ed; ERTA 2018) Line Pipe
API Spec 6D	(June 2018, 4th Ed; Errata 1 July 2018; Errata 2 August 2018) Specification for Pipeline and Piping Valves
API Spec 6FA	(1999; R 2006; Errata 2006; Errata 2008; R 2011) Specification for Fire Test for Valves
API Std 607	(2016) Fire Test for Quarter-turn Valves and Valves Equipped with Non-metallic Seats

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME B1.1	(2003; R 2018) Unified Inch Screw Threads (UN and UNR Thread Form)
ASME B16.5	(2020) Pipe Flanges and Flanged Fittings NPS 1/2 Through NPS 24 Metric/Inch Standard
ASME B16.9	(2018) Factory-Made Wrought Buttwelding Fittings
ASME B16.11	(2016) Forged Fittings, Socket-Welding and Threaded
ASME B16.21	(2021) Nonmetallic Flat Gaskets for Pipe Flanges
ASME B18.2.1	(2012; Errata 2013) Square and Hex Bolts and Screws (Inch Series)
ASME B18.2.2	(2022) Nuts for General Applications: Machine Screw Nuts, and Hex, Square, Hex Flange, and Coupling Nuts (Inch Series)
ASME B31.3	(2020) Process Piping
ASME BPVC SEC VIII D1	(2019) BPVC Section VIII-Rules for Construction of Pressure Vessels Division 1
ASME B40.100	(2013) Pressure Gauges and Gauge Attachments

ASTM INTERNATIONAL (ASTM)

ASTM A36/A36M	(2019) Standard Specification for Carbon Structural Steel
ASTM A53/A53M	(2020) Standard Specification for Pipe, Steel, Black and Hot-Dipped, Zinc-Coated, Welded and Seamless

ASTM A105/A105M	(2021) Standard Specification for Carbon Steel Forgings for Piping Applications
ASTM A193/A193M	(2020) Standard Specification for Alloy-Steel and Stainless Steel Bolting Materials for High-Temperature Service and Other Special Purpose Applications
ASTM A194/A194M	(2020a) Standard Specification for Carbon Steel, Alloy Steel, and Stainless Steel Nuts for Bolts for High-Pressure or High-Temperature Service, or Both
ASTM A234/A234M	(2019) Standard Specification for Piping Fittings of Wrought Carbon Steel and Alloy Steel for Moderate and High Temperature Service
ASTM A269/A269M	(2015; R 2019) Standard Specification for Seamless and Welded Austenitic Stainless Steel Tubing for General Service
ASTM A961/A961M	(2021) Standard Specification for Common Requirements for Steel Flanges, Forged Fittings, Valves, and Parts for Piping Applications
ASTM B696	(2000; R 2015) Standard Specification for Coatings of Cadmium Mechanically Deposited
ASTM B766	(1986; R 2015) Standard Specification for Electrodeposited Coatings of Cadmium
ASTM D229	(2019) Standard Test Methods for Rigid Sheet and Plate Materials Used for Electrical Insulation
ASTM D1418	(2010; R 2016) Standard Practice for Rubber and Rubber Lattices - Nomenclature
ASTM D3308	(2012; R 2017) Standard Specification for PTFE Resin Skived Tape
ASTM F336	(2002; R 2016) Standard Practice for Design and Construction of Nonmetallic Enveloped Gaskets for Corrosive Service
ASTM F436	(2011) Hardened Steel Washers

INSTITUTE OF ELECTRICAL AND ELECTRONICS ENGINEERS (IEEE)

IEEE 142	(2007; Errata 2014) Recommended Practice for Grounding of Industrial and Commercial Power Systems - IEEE Green Book
IEEE 1100	(2005) Emerald Book IEEE Recommended Practice for Powering and Grounding Electronic Equipment

IEEE C62.41.2 (2002) Recommended Practice on
Characterization of Surges in Low-Voltage
(1000 V and Less) AC Power Circuits

INTERNATIONAL ORGANIZATION FOR STANDARDIZATION (ISO)

ISO 1629 (2013) Rubber and Latexes - Nomenclature

MANUFACTURERS STANDARDIZATION SOCIETY OF THE VALVE AND FITTINGS
INDUSTRY (MSS)

MSS SP-58 (2018) Pipe Hangers and Supports -
Materials, Design and Manufacture,
Selection, Application, and Installation

NACE INTERNATIONAL (NACE)

NACE SP0188 (1999; R 2006) Discontinuity (Holiday)
Testing of New Protective Coatings on
Conductive Substrates

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 30 (2021; TIA 20-1; TIA 20-2) Flammable and
Combustible Liquids Code

NFPA 70 (2020; ERTA 20-1 2020; ERTA 20-2 2020; TIA
20-1; TIA 20-2; TIA 20-3; TIA 20-4)
National Electrical Code

NFPA 77 (2014) Recommended Practice on Static
Electricity

NFPA 407 (2022) Standard for Aircraft Fuel Servicing

NFPA 780 (2020) Standard for the Installation of
Lightning Protection Systems

SOCIETY FOR PROTECTIVE COATINGS (SSPC)

SSPC SP 5/NACE No. 1 (2007) White Metal Blast Cleaning

SOCIETY OF AUTOMOTIVE ENGINEERS INTERNATIONAL (SAE)

SAE AMS3275 (2009; Rev C) Sheet, Acrylonitrile
Butadiene (NBR) Rubber and Non-Asbestos
Fiber Fuel and Oil Resistant

SAE J514 (2012) Hydraulic Tube Fittings

U.S. DEPARTMENT OF DEFENSE (DOD)

MIL-STD-161 (2005; Rev G; Notice 1 2010)
Identification Methods for Bulk Petroleum
Products Systems Including Hydrocarbon
Missile Fuels

MIL-STD-3004-1 (2018) Quality Assurance for Bulk Fuels,

Lubricants and Related Products

1.3 ADMINISTRATIVE REQUIREMENTS

Design conditions must be as specified. Refer to Section 01 78 23
OPERATION MAINTENANCE DATA.

1.4 SUBMITTALS

Government approval is required for submittals with a "G" classification.
Submittals not having a "G" classification are for Contractor Quality
Control approval. Submit the following in accordance with Section 01 33 00
SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Work Plan; G

Pigging Plan; G

Hydrotesting Plan; G

SD-02 Shop Drawings

Grounding and Bonding

Pipe Supports; G

Pigging System Components; G

SD-03 Product Data

Carbon Steel Piping; G

Stainless Steel Piping; G

External Protective Coatings For Aboveground Piping; G

External Protective Coatings For Buried Steel Piping; G

Fittings; G

Carbon Steel Fittings; G

Insulating Flange Kits; G

Lightning Surge Arrester; G

Bolts, Nuts And Washers; G

Ball Valves; G

Plug (Double Block and Bleed) Valves; G

Electric Valve Actuator

Thermal Relief Valve; G

Mechanically Adjustable Segmented Elastomeric Seal; G

Pipe Sleeves; G

Sample Connections; G

Buried Utility Tape; G

Pipeline Markers; G

Flange Gaskets, Non-Metallic, Non-Electrically Isolating; G

SD-06 Test Reports

Exterior Coating Holiday Test

Pneumatic Test

Hydrostatic Test

Performance Testing

Pipe Pigging Verification

SD-07 Certificates

Contractor Qualifications; G

Pipeline Inventory; G

Survey Final Elevations; G

SD-10 Operation and Maintenance Data

Insulating Flange Kits; G

Lightning Surge Arrester; G

Ball Valves; G

Plug (Double Block and Bleed) Valves; G

Electric Valve Actuator; G

Thermal Relief Valve; G

Mechanically Adjustable Segmented Elastomeric Seal; G

Sample Connections; G

1.5 QUALITY ASSURANCE

1.5.1 Contractor Qualifications

Each installation Contractor must have successfully completed at least 3 projects of the similar scope and the same size or larger within the last 6 years. Each installation Contractor must demonstrate specific installation experience in regard to the specific system installation to be performed. Each installation Contractor must have taken, if applicable, manufacturer's training courses on the installation of piping

and must meet the licensing requirements in the state. Experience must include the erection of piping systems in compliance with the requirements of ASME B31.3, and NFPA 30. Submit a letter listing prior projects, the date of construction, a point of contact for each prior project, the scope of work of each prior project, and a detailed list of work performed. Provide in the letter evidence of prior manufacturer's training and state licensing.

1.5.2 Work Plan

Submit a comprehensive work plan that provides sufficient detail to demonstrate a thorough understanding of the project. Document that all components to be provided will function together and produce the result expected by the Government. Include any proposed dates for piping system shutdowns as well as the Contractor's ability to complete the work within the allotted shutdown periods. Show proposed dates and nature of piping system operations required of the Government. Include a list of manpower, spare piping and system component that will be on hand for each phase of the work. Describe, in detail, the means of:

- a. Coordinating work with Government and third parties.
- b. Preparing for safe piping repair work.
- c. Pneumatic pressure testing new piping sections.
- d. Hydrostatic pressure testing new piping sections.
- e. Interrupting or isolating an existing fuel service or system.
- f. Purging piping.
- g. Vapor monitoring.
- h. Preparations for containing and disposing of residual fuel.
- i. Cutting, sealing, and welding into existing piping systems.
- j. Welding tie-ins in place.
- k. Examining repair section tie-in welds.
- l. Collecting, storing and disposing of waste fuel generated during work.

1.5.3 Pigging Plan

The Contractor must submit a detailed written plan covering all aspects of the pipeline pigging operation, including anticipated pig runs, types of pigs, sequence of work, and retrieval/repair procedures. The Contractor must identify for each pigging evolution, the characteristics of the pig (type/purpose) and method/medium for propulsion.

1.5.4 Hydrotesting Plan

The Contractor must submit a detailed written plan covering all aspects of the pipeline hydrostatic testing operations, including procedures and sequencing of testing, segments of piping to be tested, fluid to be used during testing, equipment removal and isolation, hydrostatic testing pressures, safety protocols, etc.

1.5.5 Design Data

1.5.5.1 Pipeline Inventory

Fuel system volume must be calculated using as constructed pipe lengths, internal diameters, fittings, and components. Totals must be provided for all items containing fuel with the exception of tanks which is covered by other specifications. A certified pipeline inventory - a detailed list with sizes, lengths, quantity, and volumes must be provided for the systems in this project. Such systems include, but are not limited to: marine receipt, pipeline receipt, truck off-loading receipt, pump house, pump pad, truck loading, marine loading, transfer pipeline, product recovery, and other miscellaneous piping systems.

1.5.6 Delivery, Storage, and Handling

If fuel is used for hydrotesting or cleaning pigging, comply with all the requirements in Section 33 08 55 FUEL DISTRIBUTION SYSTEM START-UP (NON-HYDRANT).

PART 2 PRODUCTS

2.1 ELECTRICAL WORK

2.1.1 General

Motors, manual or automatic motor control system components except where installed in motor control centers, and protective or signal devices required for the operation specified herein must be provided under this section in accordance with Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM. Any wiring required for the operation specified herein, but not shown on the electrical plans, must be provided under this section in accordance with Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM.

2.1.2 Grounding and Bonding

Ground and bond as indicated on the drawings and in accordance with NFPA 70, NFPA 77, NFPA 407, NFPA 780, API RP 540, API RP 2003, IEEE 142, and IEEE 1100. Provide jumpers to overcome the insulating effects of gaskets, paints, or nonmetallic components.

2.2 MATERIALS

Pipe and fittings in contact with fuel must be stainless steel, interior epoxy coated carbon steel, or interior uncoated carbon steel. Fuel piping must be interior uncoated carbon steel, unless noted otherwise. No zinc coated metals, brass, bronze or other copper bearing alloys must be used in contact with the fuel. All carbon steel and stainless steel underground piping must have an exterior protective coating and must be cathodically protected in accordance with Section 26 42 17 IMPRESSED CURRENT CATHODIC PROTECTION (ICCP) SYSTEM. Cathodic protection for metal components that attach to a tank must be coordinated and compatible with the tank corrosion control system. Identification of piping must be in accordance with MIL-STD-161 unless specified otherwise. Material for manual valves must be as specified hereinafter. Do not use aluminum valves.

2.2.1 Types of Fuel

Components must be suitable for use with F-76 fuel.

2.2.2 Carbon Steel Piping

Subject each length of pipe to factory hydrostatic testing and ultrasonic testing in accordance with their respective pipe specification.

- a. Piping 300 mm and Larger: API Spec 5L Product Specification Level (PSL) 1, Grade B, seamless; or ASTM A53/A53M Grade B, seamless; all having a wall thickness of 9 mm.
- b. Piping 65 through 250 mm: Schedule 40, API Spec 5L Product Specification Level (PSL) 1, Grade B, seamless Grade B; or Schedule 40, Seamless, ASTM A53/A53M Grade B.
- c. Piping 50 mm and Smaller: Schedule 80, API Spec 5L Product Specification Level (PSL) 1, Grade B, seamless Grade B; or Seamless, Schedule 80 ASTM A53/A53M Grade B.

2.2.3 Stainless Steel Piping

- a. Stainless Steel Control Tubing: Seamless, fully annealed tubing conforming to ASTM A269/A269M, Grade TP316, Rockwell hardness B80 or less. Wall thickness for 13 mm tubing to be a minimum of 1.2 mm.

2.2.4 External Protective Coatings for Aboveground Piping

Provide exterior coating of aboveground piping and fittings, piping in pits, pipe supports, filter separators, and miscellaneous metal and system components in accordance with Section 09 97 13.27 HIGH PERFORMANCE COATING FOR STEEL STRUCTURES. Color of finish coat must be white. Do not coat aboveground stainless steel or aluminum surfaces.

2.2.5 External Protective Coatings for Buried Steel Piping

2.2.5.1 Carbon Steel Piping

- a. New pipe and fittings must be factory coated in accordance with Section 33 52 80 LIQUID FUELS PIPELINE COATING SYSTEM.
- b. Field joints and repairs must be in accordance with Section 33 52 80 LIQUID FUELS PIPELINE COATING SYSTEMS.
- c. Field joints and repairs in tight spots (valve pits when heaters are too big) must be liquid epoxy in accordance with Section 33 52 80 LIQUID FUELS PIPELINE COATING SYSTEMS.
- d. Existing systems must match existing coating system and must be in accordance with Section 33 52 80 LIQUID FUELS PIPELINE COATING SYSTEMS.

2.2.6 Fittings

2.2.6.1 General

Welding ells, caps, tees, reducers, must be of materials compatible for welding to the pipeline in which they are installed, and wall thickness, pressure and temperature ratings of the fittings must be not less than the

adjoining pipeline. Unless otherwise specified herein or required by the conditions of installation, all elbows must be the 1.5 diameter (D) type. Miter joints are not acceptable. Make odd angle offsets with pipe bends or elbows cut to the proper angle. Butt weld fittings must be factory-made wrought fittings manufactured by forging or shaping. Fabricated fittings will not be permitted. Welding branch fittings must be insert type suitable for radiographic inspections specified herein.

Make branch connections with butt-welded tees except where the branch is at least two pipe sizes smaller than the run, in which case the branch connection can be made with a forged or seamless branch outlet fitting. The branch outlet fitting must be designed in such a way that the connection can be radiographed. The branch outlet fittings may be a non-radiographicable if: the piping it is connected to is aboveground, the branch outlet size is 2.5 inches or less in diameter, and the branch outlet is located in contained pumphouses, contained truck offloads, contained truck fill stands, and other visibly contained areas equipped with containment curb.

2.2.6.2 Carbon Steel Fittings

- a. Fittings 65 mm and Larger: Butt weld, conforming to ASTM A234/A234M, grade WPB and ASME B16.9 of the same wall thickness as the adjoining pipe. All welds on buried piping must be radiographically examined throughout the entire length of each weld. Each fitting must be subjected to the Supplementary Requirements S3 and S4, Liquid Penetration examination and Magnetic-Particle Examination. Detectable flaws will not be accepted in the supplementary examinations. Fittings must be identified to relate them to their respective radiograph.
- b. Fittings 50 mm and Smaller: Forged, butt weld or socket welded (except flanges, see below). If specifically indicated on drawings, non-flange fittings may be threaded. Socket welded or threaded fittings must be Class 3000, conforming to ASTM A105/A105M, Grade 2 and ASME B16.11. Threaded fittings must only be used for above grade applications. Underground and in pits, low point drain pipe, and high point vent pipe must be butt welded.
- c. Flanges: Class 150 weld neck, butt weld, forged flanges conforming to ASTM A105/A105M, and ASME B16.5 except flanges that are to be connected to pumps must match the pump flanges rating. Threaded and slip-on flanges are not allowed. Flanges to be 2 mm raised face with modified spiral serrated gasket surface finish, except where required otherwise to match system components furnished. Match flange face to valves or system components furnished. Flange face must be machined to match valves or system components furnished. Use of spacing rings or gaskets discs are not allowed. Flanges must be subjected to the Supplementary Requirements S56, Liquid Penetrant Examination as outlined in ASTM A961/A961M. Detectable flaws will not be accepted. For flanges 50 mm and smaller located in contained pumphouses, contained truck offloads, contained truck fill stands, and other visibly contained areas the fitting may be forged (socket welded), Class 150, conforming to ASTM A105/A105M, Grade 2 and ASME B16.5. In pits, vaults, on thermal relief valve piping for pipeline routes, and other uncontained locations the flanges must be radiographicable, butt welded, weld neck type.
- d. Piggable System

(1) Provide barred tees on all branch outlets **50 mm** and larger when within **6.1 meters** of pig launcher or receiver barrel, including the barrels. Provide barred tees on all size outlets greater than **50 mm** in size with any part of the outlet on the bottom half of the pipe. Provide barred tees on all branch connections equal to or greater than 50 percent of piggable line size.

(2) Use 3 D sweeps between pig launchers and receivers.

2.2.6.3 Stainless Steel Fittings

- a. Stainless Steel Tube Fittings. Flareless, 316 stainless steel fittings conforming to **SAE J514**.

2.2.7 Insulating Flange Kits (Electrically Isolating)

Provide **ASTM D229** electrical insulating material of 1,000 ohms minimum resistance or 500 Volts per mil (VPM) minimum dielectric strength; material must be resistant to the effects of aviation and non-aviation hydrocarbon fuels. Provide full face insulating gaskets between flanges. Provide full surface **0.75 mm** thick wall thickness, spiral-wound mylar insulating sleeves between the bolts and the holes in flanges; bolts may have reduced shanks of a diameter not less than the diameter at the root of threads. Provide **3 mm** thick high-strength phenolic insulating washers next to flanges and provide flat circular stainless steel washers over insulating washers and under bolt heads and nuts. Provide bolts **12 mm** longer than standard length to compensate for the thicker insulating gaskets and the washers under bolt heads and nuts. Above grade flanges separated by electrically insulating flange kits must be provided with weatherproof **lightning surge arrester** devices. The surge arrester must bolt across flanges separated by insulating gasket kits per detail on contract drawings. Provide with flange protector as described in this section. The arrester must have the following features:

- a. Weatherproof NEMA 4 enclosure.
- b. Bidirectional and bipolar protection.
- c. Constructed of solid state components, no lights, fuses or relays and used without required maintenance or replacement.
- d. Withstand unlimited number of surges at 50,000 Amperes.
- e. Maximum clamping voltage of 700 Volts based on a **IEEE C62.41.2** 8x20 microsecond wave form at 50,000 Amperes peak measured at the device terminals (zero lead length).
- f. A UL listed arrester for installation in Class 1, Division 1 and Division 2, Group D, hazardous areas.

Install the mounting bracket and leads on the flange side of the bolt insulating sleeve and washer, and size in accordance with this schedule:

Line Size	Bolt Size
50 mm	16 mm

Line Size	Bolt Size
65 mm	16 mm
80 mm	16 mm
100 mm	16 mm
150 mm	19 mm
200 mm	19 mm
250 mm	22 mm
300 mm	22 mm
350 mm	25 mm
400 mm	25 mm
Note: Make allowance for the 1 mm thickness of the insulating sleeve around the bolts when sizing the mounting lugs.	

2.2.8 Bolts, Nuts and Washers

- a. Bolts and nuts for pipe flanges, flanged fittings, valves and accessories must conform to ASME B18.2.1 and ASME B18.2.2, except as otherwise specified.
- b. Bolts must be of sufficient length to obtain full bearing on the nuts and must project no more than three full threads and no less than two full threads beyond the nuts with the bolts tightened to the required torque.
- c. Bolts must be regular hexagonal bolts conforming to ASME B18.2.1 with material conforming to ASTM A193/A193M, Class 2, Grade B8, stainless steel, when connections are made where a stainless steel flange is involved, and Grade B7, chromium molybdenum alloy, when only carbon steel flanges are involved. Bolts and nuts chosen must have sufficient strength to seat gasket types chosen. Bolts must be threaded in accordance with ASME B1.1, Class 2A fit, Coarse Thread Series, for sizes 25 mm and smaller and Eight-Pitch Thread Series for sizes larger than 25 mm.
- d. Nuts must conform to ASME B18.2.2, hexagonal, heavy series with material conforming to ASTM A194/A194M, Grade 8, stainless steel for stainless steel bolts, and Grade 7, chromium molybdenum alloy for chromium molybdenum alloy bolts. Nuts must be threaded in accordance with ASME B1.1, Class 2B fit, Coarse Thread Series for sizes 25 mm and smaller and Eight-Pitch Thread Series for sizes larger than 25 mm.
- e. Provide washers under bolt heads and nuts. Use chromium molybdenum alloy washers dimensioned to ASTM F436 flat circular for chromium molybdenum bolts. Stainless steel washer dimensioned similar to ASTM F436 flat circular, use material the same as the bolt.
- f. Use torque wrenches to tighten all flange bolts to the torque recommended by the gasket manufacturer. Tight in the pattern recommended by the gasket manufacturer. Use anti-seize compound on stainless steel bolts.

2.2.9 Flange Gaskets, Non-Metallic, Non-Electrically Isolating

ASME B16.21, composition ring, using a Nitrile Rubber such as Buna-N and NBR, polytetrafluoroethylene (PTFE), or a fluoro rubber such as FKM, FPM and Viton®. The gasket must be 3 mm thick. Gaskets must be resistant to the effects of aviation and non-aviation hydrocarbon fuels and manufactured of fire-resistant materials. Full-face gaskets must be used for flat-face flanged joints. Ring gaskets must be used for raised-face flanged joints. Gaskets must be of one piece factory cut. Select a gasket suitable for the working and test pressure of the fluid.

2.2.9.1 Nitrile Butadiene (Buna-N)

Provide Buna-N material that conforms to SAE AMS3275.

2.2.9.2 Acrylonitrile Butadiene Rubber (NBR)

Provide NBR material that conforms to SAE AMS3275.

2.2.9.3 Polytetrafluoroethylene (PTFE)

Provide PTFE material that conforms to ASTM F336.

2.2.9.4 Fluoro Rubber FKM

Provide FKM material that conforms to ASTM D1418.

2.2.9.5 Fluoroelastomer FPM

Provide FPM material that conforms to ISO 1629.

2.3 MANUAL VALVES

All portions of a valve coming in contact with fuel in stainless steel pipelines or epoxy lined carbon steel pipelines must be of noncorrosive material. Valves in stainless steel pipelines or epoxy lined carbon steel pipelines must be Type 304 or Type 316 stainless steel or carbon steel internally plated with chromium or nickel or internally electroless nickel plated. Valves in unlined carbon steel pipelines must have carbon steel body. Stem and trim must be stainless steel for all valves. Manually operated valves 150 mm and larger must be worm-gear operated and valves smaller than 150 mm must be lever operated or handwheel operated. Valves smaller than 50 mm must have lever-type handles. Valves installed more than 2.4 m above finished floor must have chain operators and a position indicators visible from ground level. Sprocket wheel for chain operator must be aluminum. Valves in the piggable line flow path between the pig launchers and the pig receivers, including the valves in the isolation valve pits must be full bore, piggable, double block and bleed type. The full bore piggable valves at the launcher and the receiver must be ball type. Valves must be true full bore with no projections extending into the flow path of the pig train.

2.3.1 Ball Valves

Ball valves must be fire tested and qualified in accordance with the requirements of API Std 607 and API STD 608. Seal material for the fire test must be graphite, seal material for the project must be as indicated below. Ball valves must be nonlubricated valves that operate from fully open to fully closed with 90-degree rotation of the ball.

Valves 2 inches and larger must conform to applicable construction and dimension requirements of **API Spec 6D**, ANSI Class 150 and must have flanged ends. Valves **50 mm** and larger must conform to applicable construction and dimension requirements of **API Spec 6D**, ANSI Class 150 and must have flanged ends. The balls in valves **250 mm** and larger full port and **300 mm** and larger regular port and larger must have trunnion type support bearings. Except as otherwise specified or indicated, reduced port or full port valves may be provided at the Contractor's option. Balls must be solid, not hollow cavity.

2.3.1.1 Materials

Ball must be stainless steel. Ball valves must have tetrafluoroethylene (TFE) or fluoroelastetomer (FKM), commonly referred to as Viton seats, body seals and stem seals. Valves **100 mm** and smaller must have a locking mechanism.

2.3.2 Plug (Double Block and Bleed) Valves

API Spec 6D, **API Spec 6FA**, ANSI Class 150, non-lubricated, resilient, double seated, trunnion mounted, tapered lift plug capable of two-way shutoff. Valve must have tapered plug of steel or ductile iron with chrome or nickel plating and plug supported on upper and lower trunnions. Sealing slips must be steel or ductile iron, with Viton seals which are held in place by dovetail connections. Valve design must permit sealing slips to be replaced from the bottom with the valve mounted in the piping. Valves must operate from fully open to fully closed by rotation of the handwheel to lift and turn the plug. Valves must have weatherproof operators with mechanical position indicators. Indicator shaft must be stainless steel. Minimum bore size must be not less than 65 percent of the internal cross sectional area of a pipe of the same nominal diameter unless bore height of plug equals the nominal pipe diameter and manufacturer can show equal or better flow characteristics of the reduced bore size design. Valves in the piggable line flow path between the pig launchers and the pig receivers, including the valves in the isolation valve pits, in fuel piping between the pig launchers and the pig receivers must be full bore, piggable. Valves must be true full bore with no projections extending into the flow path of the pig train. Full port plug valves in distribution piping must be provided with a **15 mm** threaded body drain.

2.3.2.1 General

Valves in the operating tank suction and fill lines and the valves at the four valve manifold in the pump room in the tank fill lines must be provided with a factory-installed limit switch that is actuated by the valve closure. Tank fill line valve and four valve manifold limit switches must be provided with one double pole double throw contacts or four single pole, double throw contracts, two for open, two for closed. Tank suction line valve limit switches must be provided with one double pole double throw contacts or four single pole, double throw contacts, for closed, and one single pole double throw contact or two single pole, double throw contacts for open. All components must be watertight and U.L. listed for Class I, Division 1, Group D hazardous areas.

2.3.2.2 Valve Operation

Rotation of the handwheel toward open must lift the plug without wiping the seals and retract the sealing slips so that during rotation of the

plug clearance is maintained between the sealing slips and the valve body. Rotation of the handwheel toward closed must lower the plug after the sealing slips are aligned with the valve body and force the sealing slips against the valve body for positive closure. When valve is closed, the slips must form a secondary fire-safe metal-to-metal seat on both sides of the resilient seal. Plug valves located in Isolation Valve Pits or vaults must be provided with handwheel extensions.

2.3.2.3 Relief Valves

ANSI Class 150. Provide plug valves with automatic thermal relief valves to relieve the pressure build up in the internal body cavity when the plug valve is closed. Relief valves must open at 175 kPa differential pressure and must discharge to the throat of, and to the upstream side, of the plug valve.

2.3.2.4 Bleed Valves

ANSI Class 150, stainless steel body valve. Provide manually operated bleed valves that can be opened to verify that the plug valves are not leaking when in the closed position.

2.3.2.5 Electric Valve Actuator

The actuator, controls and accessories must be the responsibility of the valve-actuator supplier for sizing, assembly, certification, field-testing and any adjustments necessary to operate the valve as specified. The electric valve actuator must include as an integral unit the electric motor, actuator unit gearing, limit switch gearing, position limit switches, torque switches, drive bushing or stem nut, declutch lever, wiring terminals for power, remote control, indication connections and handwheel. The electrically actuated plug valve must be set to open and close completely in 30 to 60 seconds against a differential pressure of 2 MPa. The actuator settings of torque and limit contacts must be adjustable. The valve actuator must be suitable for mounting in a vertical or horizontal position and be rated for 30 starts per hour. The valve actuator must be capable of functioning in an ambient environment temperature.

- a. The electrical enclosure must be specifically approved by UL or Factory Mutual for installation in Class I, Division 1, Group D locations.
- b. The electric motor must be specifically designed for valve actuator service and must be totally enclosed, non-ventilated construction. The motor must be capable of complete operation at plus/minus 10 percent of specified voltage. Motor insulation must be a minimum NEMA Class F. The motor must be a removable subassembly to allow for motor or gear ratio changes as dictated by system operational requirements. The motor must be equipped with an embedded thermostat to protect against motor overload and also be equipped with space heaters. It must de-energize when encountering a jammed valve.
- c. The reversing starter, control transformer and local controls must be integral with the valve actuator and suitably housed to prevent breathing or condensation buildup. The electromechanical starter must be suitable for 30 starts per hour. The windings must have short circuit and overload protection. A transformer, if needed, must be provided to supply all internal circuits with 24 VDC or 110 VAC may be

used for remote controls.

- d. The actuator gearing must be totally enclosed in an oil-filled or grease-filled gearcase. Standard gear oil or grease must be used to lubricate the gearcase.
- e. The actuator must integrally contain local controls for Open, Close and Stop and a local/remote three position selector switch: Local Control Only, Off, and Remote Control plus Local Stop Only. A metallic handwheel must be provided for emergency operation. The handwheel drive must be mechanically independent of the motor drive. The remote control capability must be to open and close. Rim pull to operate valve manually must not exceed 28 kg.
- f. Position limit switches must be functional regardless of main power failure or manual operation. Four contacts must be provided with each selectable as normally open or normally closed. The contacts must be rated at 5A, 120 VAC, 30 VDC.
- g. Each valve actuator must be connected to a PLC supplied by "others."
- h. The actuator must have a local display of position even when power has been lost.
- i. The actuator must be supplied with a start-up kit comprising installation instruction, electrical wiring diagram and spare cover screws and seals.
- j. The actuator must be performance tested and a test certificate must be supplied at no extra charge. The test should simulate a typical valve load with current, voltage, and speed measured.

2.4 THERMAL RELIEF VALVE

2.4.1 Valve Material

Valves must have carbon steel bodies (stainless steel on stainless steel pipelines) and bonnets with stainless steel springs and trim. Valves must be Class 150 flanged end connections.

2.4.2 Thermal Relief Valve (ASME Type)

Thermal relief valves must be the fully enclosed, spring loaded, angle pattern, single port, hydraulically operated type with plain caps, and must be labeled in accordance with ASME BPVC SEC VIII D1. Valve stems must be fully guided between the closed and fully opened positions. The valves must be factory-set to open at pressures indicated on the drawings. Operating pressure must be adjustable by means of an enclosed adjusting screw. The valves must have a minimum capacity of 20 GPM at 10 percent overpressure. Valves must have a replaceable seat. Relief valves that do not relieve to a zone of atmospheric pressure or tank must be a balanced type relief valve.

2.4.3 Thermal Relief Valve (Balanced Type)

Thermal relief valves that do not relieve to a zone of atmospheric pressure or atmospheric tank must be a balanced type relief or regulator valve.

Thermal relief valves must be the fully enclosed, spring loaded, angle pattern, single port, fully balanced type (back pressure must not affect relief pressure) back pressure regulator/relief valve. Set valve at pressure indicated on drawings. Valve body must have 25 mm (minimum) raised face flange connections unless otherwise indicated. Orifice must have a minimum orifice size of 15 mm in diameter. Valve must have bubble-tight piston and seat design with stainless steel piston and Viton seat. Valve must be selected for the nominal flow condition of: pass a minimum of 18 liters per minute, at a differential pressure of 758 kilopascal, with a nominal set pressure of 689 kilopascal. Valve must be factory configured to open at required set pressure but must be field adjustable by means of an enclosed adjusting screw for set pressures up to 1034 kilopascals.

2.5 PIPING ACCESSORIES

2.5.1 Mechanically Adjustable Segmented Elastomeric Seal

Mechanically adjustable segmented elastomeric seals must be constructed of fuel resistant Buna-N elastomers and Type 316 stainless steel fasteners and hardware.

2.5.2 Pipe Sleeves

Pipe sleeves must be installed where indicated and at all points where the piping passes through concrete construction. Such sleeves must be of sufficient inside diameter to provide a minimum clear distance between the pipe and the sleeve of 13 mm. Sleeves through concrete pits or slabs must be standard weight carbon steel pipe with a protective coating. Each sleeve must extend through the respective pit wall or slab and must be provided with a wrap around Buna-N end seal (boot). (Viton when exposed to sunlight) and secured to the pipe sleeve and piping with adjustable stainless steel hose clamps. Sleeves where piping passes under roads or piping indicated to be double walled must be standard weight carbon steel pipe with a protective coating as previously specified. Alignment of the sleeve and piping must be such that the pipe is accurately centered within the sleeve by a nonconductive centering element. The sleeve must be securely anchored to prevent dislocation. Closure of space between the pipe and the pipe sleeve must be by means of a mechanically adjustable segmented elastomeric seal. The seal must be installed so as to be flush.

2.5.3 Pressure Gauge

Pressure gauges must conform to ASME B40.100 with metal cases and 100 mm diameter white dials. Gauges must be bottom connected, without back flanges. A pulsation dampener, adjustable to the degree of dampening required, must be provided for each gauge. Range of gauges must be as indicated. A ball valve must be provided for each pressure gauge. Gauges must be labeled with the calibration date.

2.5.4 Differential Pressure Gauge

The gauge must consist of a spring-supported, corrosion resistant piston moving inside a glass cylinder, with high pressure applied on top of the piston and low pressure applied below it. Under a differential pressure of 206.8 kPa, leakage past the piston must not exceed 120 drops per minute. The cylinder and flanges must be stainless steel with Viton O-ring seals. The high pressure inlet of the gauge must have a 10-micron pleated paper filter and the low pressure connection must have a fine mesh

stainless steel strainer. The gauge must have an operating pressure of 2068 kPa. Differential pressure range of the gauge through approximately 75 mm of piston movement must be 0-2068 kPa with an accuracy of plus 34.5 kPa, calibrated linearly. High and low pressure connections must be 8 mm NPT female with a stainless steel bar stock valve at each connection. Construction of the gauge must be such that a 3-valve manifold is not necessary. If only one bar stock valve is closed, the gauge must not be damaged by up to 206.8 kPa differential pressure in either direction. A pressure gauge must be attached to the differential pressure gauge to indicate the high pressure and have a range of 2068 kPa.

2.5.5 Pipe Supports

2.5.5.1 General

Pipe supports must conform to [MSS SP-58](#). Design pipe supports to meet the applicable requirements of ANSI/ASME [B31.3](#) or ANSI/ASME B31.4. Provide hot-dip galvanized finish on rods, nuts, bolts, washers, and supports. Provide miscellaneous metal that conforms to [ASTM A36/A36M](#), standard mill finished structural steel shapes, hot-dipped galvanized.

2.5.5.2 Adjustable Pipe Supports

Adjustable pipe supports must consist of a cast iron saddle and a threaded nipple connected to a carbon steel pipe by means of a special reducer conforming to [MSS SP-58](#). The supports must be provided with PTFE insulation strips.

2.5.5.3 Low Friction Supports

Supports must have self-lubricating anti-friction bearing elements composed of 100 percent virgin tetrafluoroethylene polymer and reinforcing aggregates, prebonded to appropriate backing steel members. The coefficient of static friction between bearing elements must be 0.06 from initial installation for both vertical and horizontal loads and deformation must not exceed [51 micrometers](#) under allowable static loads. Bonds between material and steel must be heat cured, high temperature epoxy. Design pipe support elements for the loads applied. Provide anti-friction material with a minimum of [2.3 mm](#) thick. Provide hot-dipped galvanized steel supports. Provide supports that are factory designed and manufactured.

2.5.5.4 U-bolt Half Round Supports

Supports must have anti-friction bearing half-round in contact with the bottom of the pipe. Provide Polytetrafluoroethylene or like hydrophobic, anti-corrosive material half-round with a compressive strength of at least [69 mPa](#) or greater as required. U-bolts must be installed in either a loose or limited guide configuration as indicated on the design drawings. Provide hot-dip galvanized u-bolts with seamless non-metallic low friction coating. U-bolt connection must be double nutted on the bottom and single bolted on top.

2.5.5.5 Concrete and Grout

Concrete and grout for anchors and supports must comply with Section [03 30 00](#) CAST-IN-PLACE CONCRETE.

2.5.6 Sample Connections

- a. Sample connections must be factory assembled units specifically designed for obtaining representative samples from fuel pipelines. Each connection must include a 6 mm sampling probe where the probe faces upstream, ball valve and 6 mm quick disconnect coupling with dust plug, all assembled into a unit that is suitable for installation in a pipe nipple. The sampling probe must extend not less than 25 mm into the fuel pipe. All materials in the sample connections must be stainless steel or aluminum.
- b. Furnish two sampling hose assemblies to the Contracting Officer at the project site. Each assembly must consist of a 1.8 m length of 6 mm clear plastic tubing with internal bonding/grounding wire. One end of the tubing will contain a male connector that actuates flow when inserted into the quick disconnect coupler. Each end of the bonding/grounding wire must be equipped with clips for attaching to the pipe and metal sample container.

2.6 PIGGING SYSTEM COMPONENTS

2.6.1 Maintenance Pig Launchers and Receivers

Construct of the same materials as the pipe, valves and fittings for a Class 150 system. The length of the straight barrel and the receiver line size section must be 1.5 meters each. Provide associated launcher kicker piping and receipt bypass piping, not less than 80 mm in size for 150 mm lines and 100 mm for up to 250 mm and 150 mm for up to 350 mm pipelines.

2.6.2 Smart Pig Launchers and Receivers

Construct of the same materials as the pipe, valves and fittings for a Class 150 system. The length of the straight barrel and the line size section must each be 4.5 meter. Provide associated launcher kicker piping and receipt bypass piping, not less than 80 mm in size for 150 mm lines and 100 mm for up to 250 mm and 150 mm for up to 350 mm pipelines.

2.6.3 Launcher and Receiver Closure Door

The closure must be hinged, swing bolted closure of the same material as the pipe and for a Class 150 system. Gasket must be nitrile or viton. Eye bolts must be pinned to lugs on the hub.

2.6.4 Signaler

The pig signaler must be mechanical flag type with manual reset, and be located on the pig launcher and the pig receiver. Material in contact with the fuel must be stainless steel. Units must be suitable for removal and installation under line pressure of 1.90 mPa. Signaler must be capable of withstanding line pressure of a Class 150 system.

2.7 MISCELLANEOUS ACCESSORIES

2.7.1 Concrete Anchor Bolts

Concrete anchors must conform to ASTM A307, Grade C, hot-dipped galvanized.

2.7.2 Coatings for Bolts, Studs, Nuts, and Washers

Carbon steel bolts, studs, nuts, and washers must be provided with a factory applied cadmium coating that conforms to [ASTM B696](#) or [ASTM B766](#).

2.7.3 Polytetrafluoroethylene (PTFE) Tape

Tape must conform to [ASTM D3308](#).

2.7.4 Pipe Sleeves

Provided sleeves constructed of [Coated Carbon](#) steel [piping](#), ductile iron, or cast-iron pipe conforming to [ASTM A53/A53M](#), Standard weight.

2.7.5 Buried Utility Tape

Provide detectable aluminum foil plastic-backed tape or detectable magnetic plastic tape for warning and identification of buried piping. Tape must be detectable by an electronic detection instrument. Provide tape in minimum [75 mm](#) width rolls, color coded for the utility involved, with warning identification imprinted in bold black letters continuously and repeatedly over entire tape length. Warning identification must be at least [25 mm](#) high and must state as a minimum "BURIED JET FUEL PIPING BELOW". Provide permanent code and letter coloring that is unaffected by moisture and other substances contained in trench backfill material.

2.7.6 Pipeline Markers

Provide pipeline markers constructed of [150 mm](#) diameter, one-half inch thick bronze disk with a [75 mm](#) long bronze headed bolt welded to the back of the disk. Engrave the front of the disk with the words "UNDERGROUND FUEL LINE" in the case of one line and "UNDERGROUND FUEL LINES" in the case of multiple fuel lines.

2.8 FINISHES

2.8.1 Factory Coating

2.8.1.1 Valves

Valve surfaces must be blasted clean according to [SSPC SP 5/NACE No. 1](#). Valve surfaces must be primed and coated in accordance with Section [09 97 13.27](#) HIGH PERFORMANCE COATING FOR STEEL STRUCTURES.

2.8.1.2 Equipment and Components

Unless otherwise specified, provide equipment and components fabricated from ferrous metal with the manufacturer's standard factory finish. For equipment and component surfaces subject to temperatures above [50 degrees C](#), the factory coating must be appropriately designed for the temperature service.

2.8.2 Field Painting

Painting required for surfaces not otherwise specified must be field painted as specified in Section [09 97 13.27](#) HIGH PERFORMANCE COATING FOR STEEL STRUCTURES. Do not paint aboveground stainless steel and aluminum surfaces. Do not coat equipment or components provided with a complete factory coating. Hot dip galvanized pipe support steel must not receive

the Zinc-Rich Epoxy Primer Coat only the Epoxy Intermediate Coat and Polyurethane Topcoat must be applied to these surfaces. Prior to any field painting, clean surfaces to remove dust, dirt, rust, oil, and grease.

PART 3 EXECUTION

3.1 GENERAL

Installation, workmanship, fabrication, assembly, erection, examination, inspection, and testing must be in accordance with ASME B31.3 and NFPA 30, except as modified herein. Safety rules as specified in NFPA 30 and NFPA 407 must be strictly observed. Never direct bury threaded connections, socket welded connections, unions, flanges, valves, air vents, or drains. Install all work so that parts requiring periodic inspection, operation, maintenance, and repair are readily accessible.

3.2 VERIFICATION OF DIMENSIONS

After becoming familiar with details of the work, verify dimensions in the field, and advise the Contracting Officer of any discrepancy before performing any work.

3.3 CLEANING OF PIPING

Keep the interior and ends of all new piping, affected by construction operations, thoroughly cleaned of foreign matter and water before and after being installed. Piping systems must be kept clean during installation by means of plugs or other approved methods. When work is not in progress, open ends of piping and fittings must be closed so that no water or other foreign substance will enter the pipes or fittings. Piping must be inspected before placing into position. The interior of each length of pipe must be cleaned after welding ensuring that the interior of the piping is free of foreign matter when it is connected into the system.

3.4 TRENCHING AND BACKFILLING

Trenching and backfilling must conform to Section 31 23 00.00 20 EXCAVATION AND FILL, and the following bedding and backfill requirements. The pipe must be laid in a bed of sand 150 mm deep, compacted in accordance with Section 31 23 00.00 20 EXCAVATION AND FILL. Sand must meet the requirements of Section 31 23 00.00 20 EXCAVATION AND FILL. The full length of each section of pipe without any protective covering must be excavated to permit installation of the protective covering. Pipe that has the grade or joint disturbed after laying, must be taken up and relaid. Pipe must not be laid in water or when the trench or weather conditions are unsuitable for such work. After testing and application of protective covering to joints, sand backfill must be placed and compacted around the pipe or protective coating to a depth of 300 mm above top of pipe. The remainder of the backfill must be the same as for other types of pipe.

3.5 PIPING LAYOUT REQUIREMENTS

3.5.1 Pipe Fabrication

Fabricate piping to measurements established on the project site and position into place without springing or forcing. Make provisions for absorbing expansion and contraction without undue stress in any part of

the system. The use of flexible connectors in permanently mounted pump suction and discharge lines as a method of compensating for piping misalignment is not acceptable.

3.5.2 Interferences and Measurements

Provide offsets, fittings, and accessories required to eliminate interferences and to match actual system components connection locations and arrangements. Verify measurements before commencing work. Submit discrepancies for clarification before proceeding with the installations to the Contracting Officer.

3.5.3 Space and Access

Keep piping, control tubing, which is not detailed close to structures and columns so as to take up a minimum amount of space. Ensure that access is provided for maintenance of system components, valves and gauges.

3.5.4 Location

Do not place unions in locations that will be inaccessible after the completion of the work. Place unions on each side of equipment.

3.5.5 Pipe Supports

Where provided hot dip galvanized pipe support steel must be surface prepped to SSPC SP 16, Brush-Off Blast Cleaning of Coated and Uncoated Galvanized Steel, Stainless Steels, and Non-Ferrous Metals. Once the steel is prepped, apply only the Epoxy Intermediate Coat and Polyurethane Topcoat per manufacturer's instructions.

Provide pipe supports with the maximum spacing as defined in Table 1 below, unless otherwise indicated. At the indicated locations. Provide additional pipe supports at concentrated piping loads (valves).

Table 1. Maximum Support Spacing									
Nominal Pipe Size (mm)	25 mm	40 mm	50 mm	80 mm	100 mm	150 mm	200 mm	250 mm	300 mm
Maximum Support Spacing (m)	2 m	2.75 m	3 m	3.5 m	4.25 m	5 m	5.75 m	6.50 m	7 m

Provide anchors where required to absorb or transmit thrust or eliminate vibration or pulsation. Provide supports near each change of direction. Select support components which do not restrict the movement of the pipe due to thermal expansion. Space supports uniformly and arrange symmetrically.

3.5.6 Structural Support

Provide supplementary or intermediate steel or other structural members as required for transmission of loads to members forming part of the

supporting structure. Piping must not be supported from other piping.

3.5.7 Grade

Where profiles of piping lines are shown on the drawings, grade the line uniformly between changes in slope or direction. Maintain gradient to within 6 mm over the entire length of pipe. When backfilling has been completed to the top of the pipe, the pipe must be surveyed at each joint, and logged by station number. Submit to the Contracting Officer for approval the survey final elevations before backfilling can continue.

3.5.8 Size Changes

Make changes in pipe size with reducing fittings. Do not use bushings. Make branch connections with butt-welded tees except where the branch is at least two pipe sizes smaller than the run, in which case the branch connection can be made with a forged or seamless branch outlet fitting. The branch outlet fitting must be designed in such a way that the connection can be radiographed. The branch outlet fittings may be a non-radiographicable if: the piping it is connected to is aboveground, the branch outlet size is 65 mm or less in diameter, and the branch outlet is located either in a pump house or on a system component pad equipped with containment curb.

3.5.9 Direction Changes

Make changes in direction of pipes with 1.5 D fittings. Where piping is to be piggable, make changes in direction with 3 D sweeps as indicated. Provide special fittings when required. Make odd-angle offsets with pipe bends or elbows cut to the proper angle. Make changes in direction with fittings, except that bending of pipe 100 mm and smaller will be permitted, provided a pipe bender is used and wide sweep bends are formed. Mitering or notching pipe or other similar construction to form elbows or tees will not be permitted. The centerline radius of bends must not be less than 6 diameters of the pipe if a pipe bender is used.

3.5.10 Threaded End Connections

Provide threaded end connections only on piping 50 mm in nominal size or smaller and only where indicated on the drawings. Provide threaded connections with PTFE tape or equivalent thread-joint compound applied to the male threads only. Not more than three threads must show after the joint is tightened.

3.5.11 Existing Pipe Systems

No interruptions or isolation of an existing fuel handling service or system may be performed unless the actions are appropriately documented in the approved work plan. Perform initial cutting of existing fuel pipe with a multiwheel pipe cutter, using a nonflammable lubricant. After cut is made, seal interior of piping with a gas barrier plug. Refer to API RP 2009 and API RP 2200. Purge interior of piping with carbon dioxide or nitrogen prior to performing any welding process.

3.5.12 Bolted Connections

For each bolted connection of stainless steel components (e.g., pipes, piping components, valves, and system components) use stainless steel bolts or studs, nuts, and washers. For each bolted connection of carbon

steel components, use carbon steel bolts or studs, nuts, and washers. Bolts project no more than three full threads and no less than two full threads beyond the nuts with the bolts tightened to the required torque. Prior to installing nuts, apply a compatible anti-seize compound to the male threads.

3.5.13 Flanges and Unions

Provide flanged joints in each line immediately preceding the connection to system components or material requiring maintenance such as pumps, general valves, control valves, strainers, and other similar items and as indicated. Assemble flanged joints square and tight with matched flanges, gaskets, and bolts.

3.5.14 Manual Valves

Install isolation plug or ball valves on each side of each system equipment, at the midpoint of looped mains, and at any other points indicated or required for draining, isolating, or sectionalizing purpose. Install valves with stems vertically up unless otherwise indicated. Provide individual supports and anchors for each valve.

3.5.15 Air Vents

Provide 40 mm air vents at all high points and where indicated to ensure adequate venting of the piping system.

3.5.16 Drains

Provide 50 mm drains at all low points and where indicated to ensure complete drainage of the piping. Drains must be schedule 120.

3.5.17 Pipe Sleeves

Provide a pipe sleeve around any pipe that penetrates a wall, floor, or slab. Do not install sleeves in structural members except where indicated or approved. Install pipe sleeves in masonry structures at the time of the masonry construction. Sleeves must be of such size as to provide a minimum of 12 mm all-around clearance between bare pipe and the sleeve. Align sleeve and piping such that the pipe is accurately centered within the sleeve by a nonconductive centering element. Securely anchor the sleeve to prevent dislocation. Closure of the space between the pipe and the pipe sleeve must be by means of a mechanically adjustable segmented elastomeric seal. The seal must be installed so as to be flush. For wall or floor penetrations, extend each sleeve through its respective wall or floor and cut flush with each surface. Seal around sleeves that penetrate through valve or fuel related pits with a Buna-N casing seal.

3.6 STRUCTURAL ATTACHMENTS

Provide attachment to building structure concrete and masonry by cast-in concrete inserts, built-in anchors, or masonry anchor devices. Apply inserts and anchors with a safety factor not less than 5. Do not attach supports to metal decking. Construct masonry anchors for overhead applications of ferrous materials only. Structural steel brackets required to support piping, headers, and system components, but not shown, must be provided under this section. Material used for support must be as specified under Section 05 50 13 MISCELLANEOUS METAL FABRICATIONS.

3.7 WELDING

3.7.1 General

All joints, unless indicated otherwise, in carbon steel and stainless steel piping systems must be welded. Do not weld carbon steel to stainless steel. Welding of fuel pipe joints must comply with Section 33 52 23.15 POL SERVICE PIPING WELDING.

3.8 INSTALLATION

3.8.1 Precautions

Take special care to ensure that the protective coating on buried pipe is not damaged during installation and that the completed system is free of rocks, sand, dirt, water, weld slag, and foreign objects including construction debris. Take the following steps to ensure these conditions.

- a. Coated pipe must be handled only with canvas or nylon slings or padded clamps. Any coating damaged by improper handling or storage must be repaired as specified.
- b. Pipe brought to the site must be stored on blocks or horses at least 450 mm above the ground and adequately supported to prevent sagging. Padded blocks or horses must be used for coated pipe. The method and height of storing coated pipe must be in accordance with the coating manufacturer's instructions. Pipe ends must be protected and capped against weather at all times, except to accommodate immediate installation.
- c. Visual inspection must be made of the inside of each length of pipe to ensure that it is clear and clean prior to installation.
- d. The open ends of the pipe system must be closed at the end of each day's work or when work is not in progress by use of expansion plugs and must not be opened until the work is resumed.
- e. A swab, with a leather or canvas belt disc to fit the inside diameter of pipe, must be pulled through each length of pipe after welding in place.
- f. Obstruction remaining in the pipe after completion of the system must be removed at the expense of the Contractor.
- g. Plasma cutters and torches are not to be used to make penetrations in the pipe or to cut pipe.
- h. After installation and backfill of the piggable piping is complete and before fuel is put in the pipe, the pipe will be cleaned per the paragraphs CLEANING PIG RUN and WIRE BRUSH RUN in this Section.

3.8.2 Protective Coatings for Buried Piping Including Stainless Steel Piping

3.8.2.1 Application of Coating System

Application of coating system must be in accordance with 33 52 80 LIQUID FUELS PIPELINE COATING SYSTEMS. Application of coating system must be in accordance with 09 97 13.27 HIGH PERFORMANCE COATING FOR STEEL STRUCTURES.

3.8.2.2 Inspection and Testing

The condition of factory field coated piping must be the responsibility of the Contractor and all damage to the protective covering during transit and handling must be repaired at no additional cost to the Government. All field coating must be subject to approval by the Contracting Officer. The entire pipe must be inspected as specified in sub-paragraph TESTING OF PROTECTIVE COATINGS under paragraph PROTECTIVE COATINGS FOR BURIED STEEL PIPING. The inspection for holidays must be performed just prior to lowering the pipe into the ditch and every precaution must be taken during lowering and backfilling to prevent damage to the protective covering.

3.8.2.3 Damage Repair

Damaged areas of coating must be repaired as specified in the preceding paragraph for fittings.

3.9 INTERIOR EPOXY COATING

When internally epoxy lined pipe is cut, the lining must be ground back from the end a minimum of 25 mm but not more than 38 mm.

3.10 INSTALLATION OF UNDERGROUND PIPE

Underground fuel pipelines must be pitched as shown on the drawings. Where not indicated they must be pitched a minimum of 50 mm per 30.5 m. 50 mm pipe size valved drain connections must be provided at all low points and 38 mm pipe size valved outlet vent connections must be provided at all high points. Vent and drain lines must terminate in male cam-type locking end connectors with matching female dust covers and installed in pits. The pipe must have cover as shown on the drawings. Drain lines must be installed at the slopes indicated.

3.10.1 Pipe Assembly

Pipe must be strung parallel and adjacent to or above a trench. The pipe must be supported on padded skids during welding and inspection of joints. Protective coating must be inspected and repaired prior to lowering the pipe into the trench. The pipe must be lowered using only canvas or nylon slings. The sling must be dug from underneath the pipe after placements and must not be pulled from underneath the pipe while in contact with it. Care must be taken to prevent damage to the pipe, welded joints or coating and any such damage must be repaired as directed by the Contracting Officer. Pressure testing of the pipe must be done after it has been placed in final position in the trench.

3.10.2 Warning Tapes in Earth Trenches

For the purpose of early warning and identification of buried pipes outside of building walls during future trenching or other excavation, continuous identification tapes must be provided in the trench. Provide metallic core or metallic-faced, acid- and alkali-resistant, polyethylene plastic warning tape manufactured for the purpose of early warning and identification of utilities buried below the tape. Tape must be at least 80 mm in width. Color of tape must be as standard with the manufacturer with respect to the type of utility buried below the tape. Tape must have lettering at least 25 mm high with warning and identification imprinted in bold black letters continuously over the entire tape length with not less

than the following identification on the tape: BURIED JET FUEL PIPING BELOW. Tape must be installed in accordance with the printed recommendations of the tape manufacturer, as modified herein. Tapes must be buried at a depth of 150 mm from the top of the subgrade or 300 mm below the top surface of earth. Provide permanent color and printing, unaffected by moisture or soil.

3.10.3 Clearances

Install pipe to be clear of contact with other pipes, pipe sleeves, casings, reinforcing steel, conduits, cables, or other metallic structures. Where pipes cross other pipes or structures with a separation of less than 150 mm, install an insulating separator. Protect the pipe from contact with a 300 mm square by 25 mm thick bituminous-impregnated canefiber board.

3.10.4 Protective Coating

When the protective coating on pipe is damaged, the Contracting Officer must be notified and must inspect the pipe before the coating is patched. If the damage to the pipe is deeper than 1.2 mm, the damage must be repaired by welding in accordance with paragraph WELDING. If the pipe is dented, out of round or damaged to the point that welding will not make it good as new, the length of pipe must be rejected.

3.10.5 Pipeline Markers

Provide above underground fuel piping spaced every 90 meters, at tees, and at changes in direction. For sections of underground piping less than 90 meters long, place at midpoint. Provide directly above pipe for single lines and between pipes where pipes run in pairs. Provide additional marker over each mid-line fitting connections for steel reinforced flexible pipe. Cast marker into 450 mm diameter, 300 mm thick concrete plug unless it is set in an area with concrete paving in which case it must be cast into the concrete paving.

3.11 SYSTEM COMMISSIONING

Conform to Section 33 08 55 FUEL DISTRIBUTION SYSTEM START-UP (NON-HYDRANT).

3.12 TESTING

Piping must be tested by pneumatic and hydrostatic pressure. Testing must comply with applicable requirements of ASME B31.3, NFPA 30 and the requirements specified herein. Hydrostatic testing must be performed using fuel as the liquid. Pressure and hydrostatic testing must be performed only after welding inspection has been completed. Labor, materials, equipment, electricity, repairs, and retesting necessary for any of the tests required herein must be furnished by the Contractor. Provide provisions to prevent displacement of the piping during testing. Keep personnel clear of the piping during pneumatic testing. Only authorized personnel must be permitted in the area during pneumatic and hydrostatic testing. Isolate system components such as pumps, tanks, filter separators, and meters from the piping system during the testing. Do not exceed the pressure rating of any component in the piping system during the testing. Following satisfactory completion of each test, relieve the test pressure and seal the pipe immediately. Piping to be installed underground must not receive field applied exterior coatings at

the joints or be covered by backfill until the piping has passed the final pneumatic tests described herein.

3.12.1 Before Backfilling

Fuel piping must not be backfilled without completing the coating, pneumatic, weld radiograph and other piping examinations and approvals. Fuel piping in trench must not have exposed welds for longer than 30 days.

3.12.1.1 Exterior Coating Holiday Test

Following installation, test the exterior coating of direct buried piping for holidays using high-voltage spark testing in accordance with **NACE SP0188**. Repair holidays and retest to confirm holiday-free coating. Text must include all existing underground piping exposed for this project.

3.12.2 Pneumatic Test

Piping to be installed underground must not receive field applied protective covering at the joints or be covered by backfill until the piping has passed the pneumatic test described herein. To facilitate the tests, isolate various sections of the piping system and test each one separately. Where such sections terminate at flanged valve points, the line must be closed by means of blind flanges in lieu of relying on the valve. Furnish tapped flanges that can be attached to the end of the section of line being tested, and that will permit a direct connection between the piping and the air compressor or pressurizing pump. No taps in the permanent line will be permitted. Furnish all necessary equipment for testing; all gauges must be subject to testing and approval of the Contracting Officer. The air used for pneumatic testing must have a dew point of no more than **5 degrees C**. Provide dehumidifying equipment on the suction or discharge side of the air compressor used to provide air for testing. Pressurizing pump must not exceed **4.7 L/s**.

3.12.2.1 Pneumatic Test Procedure

Special safety measures, including the wearing of face mask, must be taken during testing under pressure. Only authorized personnel must be permitted in the area during testing. The pneumatic test pressure must be applied in increments. A preliminary **167 kPa** test must be applied. Maintain the pressure while soapsuds or equivalent materials are applied to the exterior of the piping. While applying the soapsuds, visually inspect the entire run of piping, including the bottom surfaces, for leaks (bubble formations). If leaks are discovered, repair the leaks accordingly and retest. Repeat process until system is leak-free at **167 kPa**. Unless otherwise directed by the Contracting Officer, all piping must **then** be tested at a pressure of **167 kPa** for not less than 2-hours, during which time there must be no drop in pressure, only pressure rises with temperature. The pressure source must be disconnected during the final test period. Any leaks revealed by the test must be repaired and the test repeated.

3.12.3 Hydrostatic Test

Upon completion of pneumatic testing and after backfilling, hydrostatically test each underground piping system with fuel at in accordance with **ASME B31.3** and **API RP 1110**, with no leakage or reduction in gauge pressure for four hours. Upon completion of pneumatic testing, hydrostatically test each aboveground piping system with fuel in

accordance with ASME B31.3 and API RP 1110, with no leakage or reduction in gauge pressure for four hours. Furnish electricity, instruments, connecting devices, and personnel for test. If fuel is used for testing, comply with all the requirements in Section 33 08 55 FUEL DISTRIBUTION SYSTEM START-UP (NON-HYDRANT). Upon completion of hydrostatic testing, perform "Soak Testing" of the piping systems per API RP 1595. Duration of the test must be a minimum of 4 days and maximum of 7 days. Defects in work must be corrected at the Contractor's expense, and the test repeated until the work is proven to be in compliance with the Contract requirements.

3.12.4 Soak Testing

Upon completion of hydrostatic testing with fuel, perform "Soak Testing" of the piping systems per API RP 1595. Duration of the test must be a minimum of 4 days and maximum of 7 days.

3.12.5 Performance Testing

After the fuel system testing is completed (including pneumatic and hydrostatic testing) the fuel system must be flushed, cleaned and performance tested as specified in Section 33 08 55 FUEL DISTRIBUTION SYSTEM START-UP (NON-HYDRANT). All control valves, both manual and automatic, must be checked for leaks (any area wetted with fuel) and proper operation and adjusted, repaired or replaced to correct any defects.

3.13 PIPE PIGGING - CLEANING

3.13.1 General

Track all pigs, using transmitter and receivers, at no less than 805 m increments, but no less than at four locations. The Contractor must prepare a contingency plan for retrieving a stuck pig and repairing any piping deformations. After pigging, plug valves must be flushed of all debris using the drain port at the bottom of the valve.

3.13.2 Use of Fuel in Cleaning Pigging

Cleaning pigging with fuel will take place after the initial receipt of fuel as per Section 33 08 55 FUEL DISTRIBUTION SYSTEM START-UP (NON-HYDRANT). Ensure that the fuel that is returned to the storage tanks during the pig runs is free of gross contamination and passes the color assessment method, and meets the requirements of MIL-STD-3004-1. Provide temporary storage tanks for the high particulate and dark color fuel that accumulates in front of and behind each pig. The contractor is responsible for cleaning the off-spec fuel in order to meet the requirements of MIL-STD-3004-1 or dispose of the off-spec fuel off-base.

3.13.3 Cleaning Pig Run

Initially, a proving pig run (foam density 32 kg/m³ should be performed to ensure the system is fully piggable. Upon completion of the successful proving pig run, the piping system must be cleaned with a standard cleaning pig. This will provide thorough cleaning of the interior of the piping system. Cleaning pig must be the bi-directional disk scraper style with steel body and replaceable polyurethane guiding and sealing disks, as well as gauge plates of 80 percent pipe diameter with 3 mm segmented aluminum fins. The pig body should include bypass nozzles and transmitter cavity. Propellant must be pressurized fuel using portable pumps. The

pig must be examined after the initial run for signs of possible pipe damage, interior slag or other adhered particles. Additional runs must be performed until the amount of collected sludge or debris is minimized, as determined by the System Supplier.

3.13.4 Wire Brush Pig Run

After the cleaning pig runs, the piping system must be cleaned with a wire brush style pig. This will remove weld slag and adhered particles from the system. Wire brush pig must be the bi-directional disk style or directional cup style with two circular steel wire brushes. The pig body should include bypass nozzles and transmitter cavity. Perform wire brush pig runs until the amount of collected weld slag or debris is minimized, as determined by the System Supplier.

3.14 PIPE PIGGING VERIFICATION

3.14.1 Geometry/Ultrasonic Tool Reports

After the system is installed and prior to performance testing, a field/preliminary report must be issued and a debrief given to Government personnel onsite on the condition of the piping system that was pigged. This must be comprised of raw data in the form of a PC download or equivalent which shows a continuous scan of each data unit output. Results of a preliminary interpretation of the data must be reported. These must include as a minimum all critical anomalies. A final report must include a description of the principle of operation, explanation of raw data, presentation of raw data, data to be clearly marked with distance traveled scale with classified anomaly location and all identifiable pipeline features, and all anomalies to be classified with locations in summary tabular form, pipe wall thickness survey, as well as the software necessary to read the data. Submittal must be in the form of digital media copied to a CD or DVD (flash drives are unacceptable).

3.14.2 Pipeline Internal Inspection Operations

3.14.2.1 General

The following pigs will be propelled through the pipeline with fuel in order to inspect the pipeline: 1.7 kg density foam swab, combination poly scraper-magnetic, stainless steel wire brush (plastic brush for internally lined piping), aluminum plate gauge, and geometry/ultrasonic tool. Tracking devices must be used on all pigs. At a minimum, the sequence of pig runs must be as follows: 1) foam swab for proving and cleaning, 2) wire brush for cleaning, 3) scraper-magnetic for cleaning, 4) aluminum plate gauge for gauging internal anomalies, 5) scraper-magnetic for cleaning, 6) wire brush for cleaning, 7) scraper-magnetic for cleaning, 8) foam swab for cleaning, (Note: the number of pig flights of each type of cleaning pigs must be determined by the amount and type of debris removed. The conclusion of the cleaning process must be when debris recovered is only that from the pigs themselves. This determination will be determined by the project's System Supplier and the Contracting Officer), 9) geometry/ultrasonic tool. The pipe wall must be continuously monitored on a real-time basis during the geometry/ultrasonic pig run. Anomalies such as patches, couplings, or flanges must also be identified, and the wall thickness given. The geometry/ultrasonic pig's technician will determine if additional runs are necessary. A permanent data set of internal inspection survey findings must be generated.

3.14.2.2 Preparatory Work

The Government will bring to the attention of the Contractor all statutes, rules and regulations relevant to the performance of the work on the site (on Government property) and will also provide the Contractor with a copy of its own site regulations (if any). Provide the pigging vendors with all-available pipeline records and drawings.

3.14.2.3 Pig Load And Launch

The pig must be loaded into the pig launcher by the Contractor. The method of loading and lodging the front pig cup into the launcher must not involve the use of uncontrolled mechanical force applied to the rear of the pig.

3.14.2.4 Pipeline Operation During Pigging

All pig runs must be made with the line packed with product. The system pumps will be used to propel the pig. The new pig traps will be used for pig launch and retrieval.

3.14.2.5 Brush and Gauging Survey

Run a brush pig at least as often as previously indicated. The brush pig must be designed and provided by the geometry/ultrasonic pig vendor. Additional runs may be required based upon the amount of debris found in the pipeline. The onsite geometry/ultrasonic pig vendor's personnel must determine if additional runs are required. Immediately following the brush pig run and immediately prior to the geometry/ultrasonic survey, run, as a minimum, a single batching pig fitted with a gauge plate equal to 90 percent of the pipeline normal inside diameter. The plate is to be a segmented aluminum disk of 3 mm thickness. The plate gauge pig must also include a tracker and tracking equipment. Track the pig assembly above ground during the operation.

3.14.2.6 Geometry/Ultrasonic Survey

After a satisfactory gauging pig run, the pipeline geometric defects must be determined by a geometry/ultrasonic tool. The geometry/ultrasonic tool must provide accuracy geometric anomaly detection, and bend radius measuring capability. The data obtained must be presented in a PC software format to allow user friendly analysis and presentation. The geometry/ultrasonic tool assembly must be capable of:

- a. Operating in hydrocarbon liquid environment, specifically diesel fuel, at a pressure of up to ANSI 300 rating.
- b. Traversing the pipeline with nominal wall thickness and possible bore restrictions down to 90 percent of nominal pipe inside diameter.
- c. Traversing the pipeline length at a speed of between 60 and 100 m/min when propelled by pumped diesel fuel. Pressure differential across pig not to exceed 34 kPa.
- d. Traversing through smooth pipe bends as small as 3D (3 pipe diameters) radius and single miter bends of up to 10 degrees change of direction.
- e. Include a tracker and tracking equipment. Track the pig assembly above ground during the operation. The battery life of the tracker

must not be less than 72-hours.

f. Manual loading into the new horizontal pig trap.

The geometry/ultrasonic tool assembly instrumentation performance must be capable of:

- a. Battery life to be minimum 18-hours at operating conditions.
- b. Principle of operation to be electronically stored geometry system.
- c. Geometry sensing to span full circumference and length of pipe, with associated distance measuring method.
- d. Geometry system must be capable of:
 - (1) positive location and identification of each geometric anomaly.
 - (2) positive location and identification of each bend.
 - (3) positive location and identification of distance marker reference points of either magnetic or electronic type placed on or above the pipe.
- e. Classification of geometric anomalies to be as minimum:
 - (1) discrimination between ovality and intrusion anomalies.
 - (2) mechanical damage such as mill defects, dents, internal gouges, and buckles.
 - (3) pipeline weld defects (such as excess weld penetration).
 - (4) geometric thickness anomalies. As a minimum, these must be reported in the following categories within the listed accuracy.
 - (a) magnitude of anomaly (plus/minus 25 mm)
 - (b) span of anomaly (plus/minus 25 mm)
 - (c) ovality (plus/minus 2.5 mm)
 - (d) span of ovality (plus/minus 25 mm)
 - (e) anomaly station (plus/minus 1:2,000)

3.14.2.7 Pipe Wall Thickness Survey

The geometry/ultrasonic tool must provide accuracy measurement of pipe wall thickness (plus/minus 0.25 mm). The data obtained must be presented in a PC software format to allow user friendly analysis and presentation.

3.14.2.8 Lost Pig

The Contractor is responsible for a lost pig, finding the pig, retrieval of the pig, and all repairs, radiographs to the pipeline system and the pig.

-- End of Section --

